



PSC 2024/25 (375AA, 9CFU)

Principles for Software Composition

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08a - Complete Partial Orders

Ingredients

a partial order (to compare elements)

order preserving functions

iterative approximations

a base case

a limit solution

Partial orders

Partially ordered set

(Poset or just PO)

a set

$(P, \sqsubseteq)$ a binary relation $\sqsubseteq \subseteq P \times P$

reflexive

$$\forall p \in P. \quad p \sqsubseteq p$$

$$p \sqsubseteq p$$

antisymmetric $\forall p, q \in P. \quad p \sqsubseteq q \wedge q \sqsubseteq p \Rightarrow p = q$

transitive $\forall p, q, r \in P. \quad p \sqsubseteq q \wedge q \sqsubseteq r \Rightarrow p \sqsubseteq r$

q
 $\uparrow$
 p

$p \sqsubseteq q$ means that p and q are **comparable** and that p is less than (or equal to) q

$p \sqsubset q$ means $p \sqsubseteq q \wedge p \neq q$

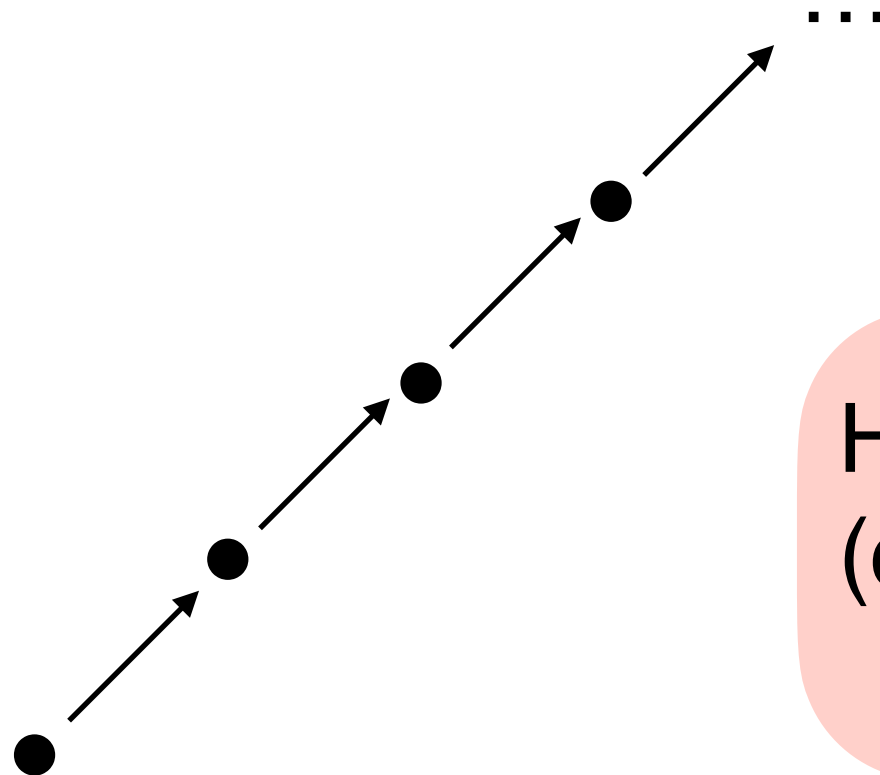
Total orders

$(P, \sqsubseteq)$ PO

total

$$\forall p, q \in P. \quad p \sqsubseteq q \vee q \sqsubseteq p$$

a PO where any two elements are **comparable**



Hasse diagram notation
(omit: reflexive arcs,
transitive arcs)

Discrete orders

$(P, \sqsubseteq)$ PO

discrete

$$\forall p, q \in P. \quad p \sqsubseteq q \Leftrightarrow p = q$$

each element is **comparable** only to itself



Flat orders

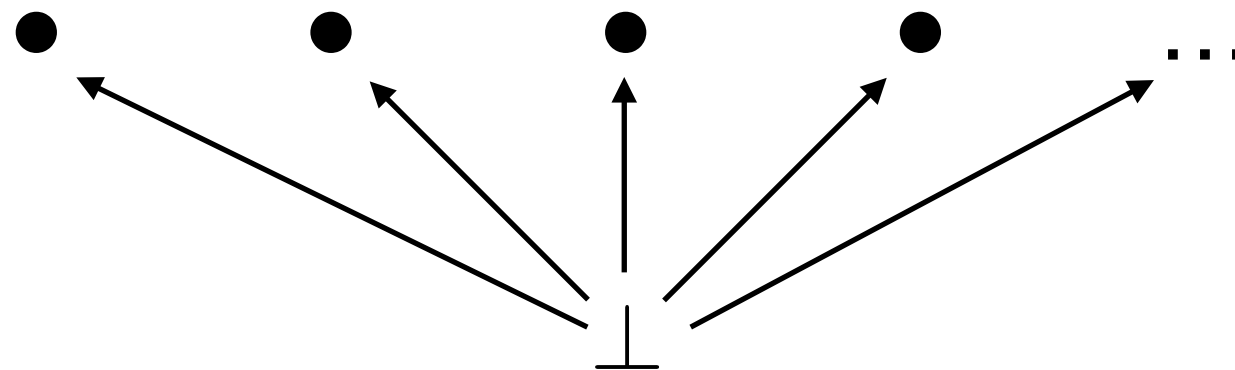
$(P, \sqsubseteq)$ PO

flat

$$\forall p, q \in P. \quad p \sqsubseteq q \Leftrightarrow p = q \vee p = \perp$$

bottom

each element is **comparable** only to itself
and with a distinguished (smaller) element $\perp$





Exercise

$(\mathbb{N}, \leq)$

PO?



Total?



Discrete?



Flat?



...



3



2



1



0



Exercise

$(\wp(S), \subseteq)$

PO?



Total?

$$|S| < 2$$

Discrete?

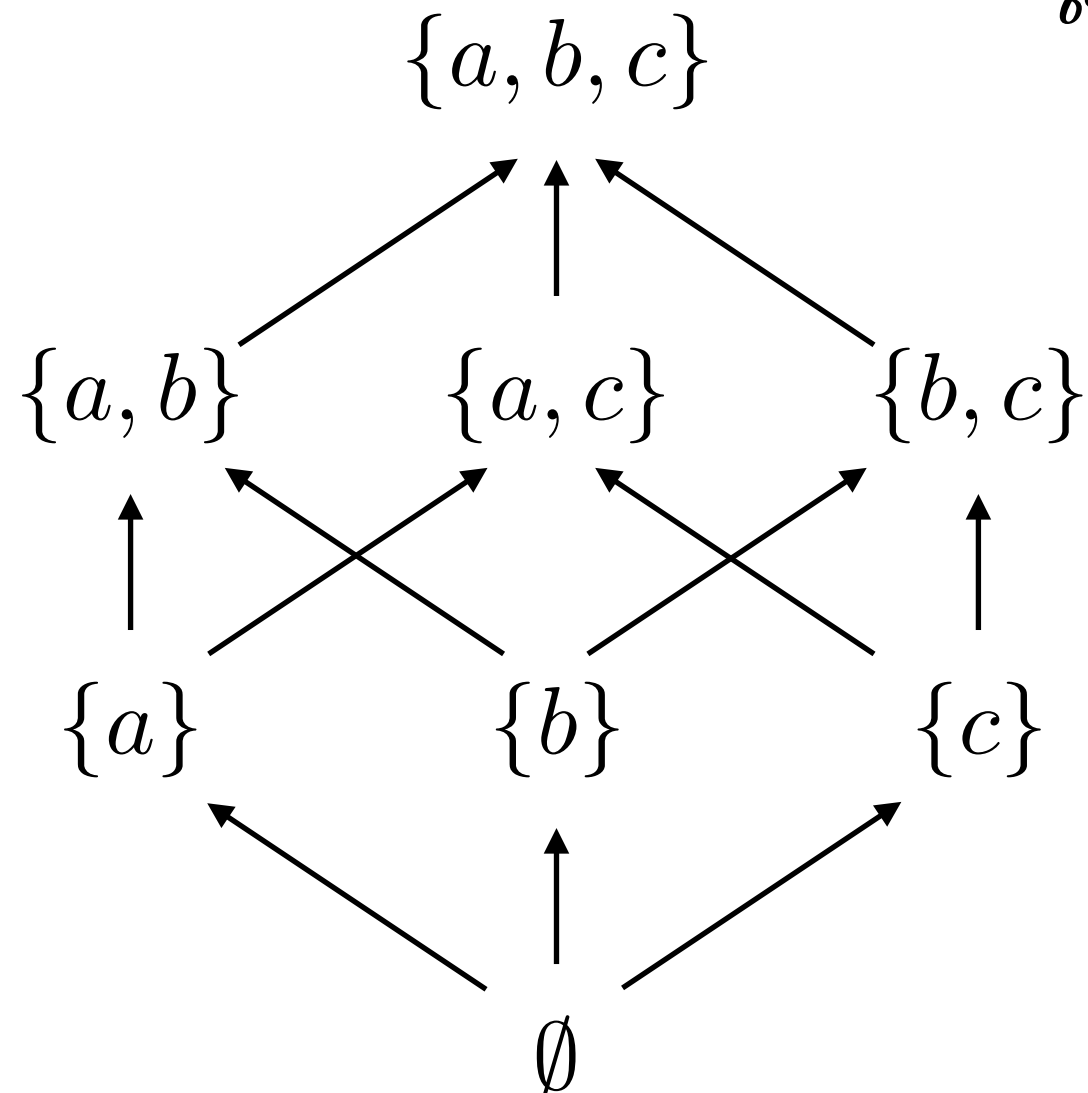
$$S = \emptyset$$

Flat?

$$|S| < 2$$

example: $S = \{a, b, c\}$

$$\begin{aligned}\wp(\emptyset) &= \{\emptyset\} \\ \wp(\{a\}) &= \{\emptyset, \{a\}\}\end{aligned}$$



$$\begin{array}{ccc}\{a, b\} & \not\subseteq & \{b, c\} \\ & \not\supseteq & \\ \{a\} & \not\subseteq & \{b\} \\ & \not\supseteq & \end{array}$$



Exercise

$(\mathbb{N}, =)$

PO?

Total?

Discrete?

Flat?



0

1

2

3

...



Exercise

$(\mathbb{N} \cup \{\perp\}, \{(\perp, n) \mid n \in \mathbb{N}\}^*)$

PO?



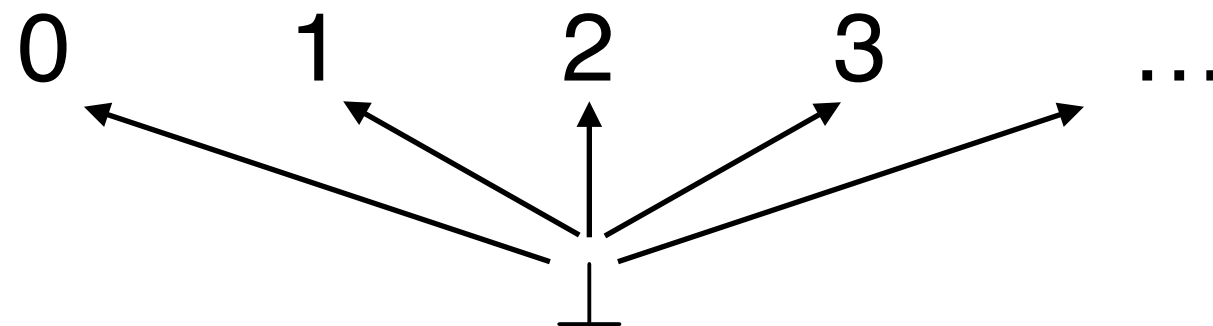
Total?



Discrete?



Flat?





Exercise

$(\mathbb{N} \cup \{\infty\}, \leq)$

PO?



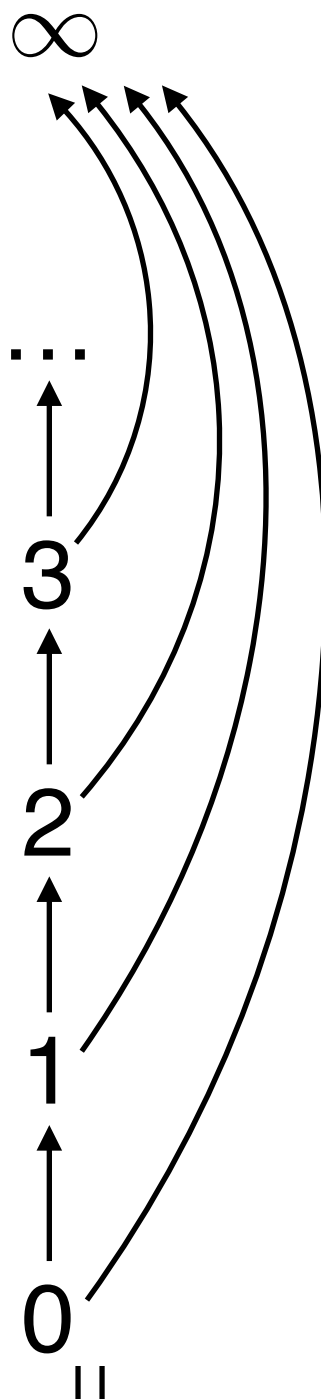
Total?



Discrete?



Flat?





Exercise

	PO?	Total?	Discrete?	Flat?
$(\mathbb{N}, <)$				
$(\mathbb{Z}, \leq)$				
$(\mathbb{Z} \cup \{-\infty, \infty\}, \leq)$				
$(T_\Sigma, \prec)$				
$(\mathbb{N}, \neq)$				

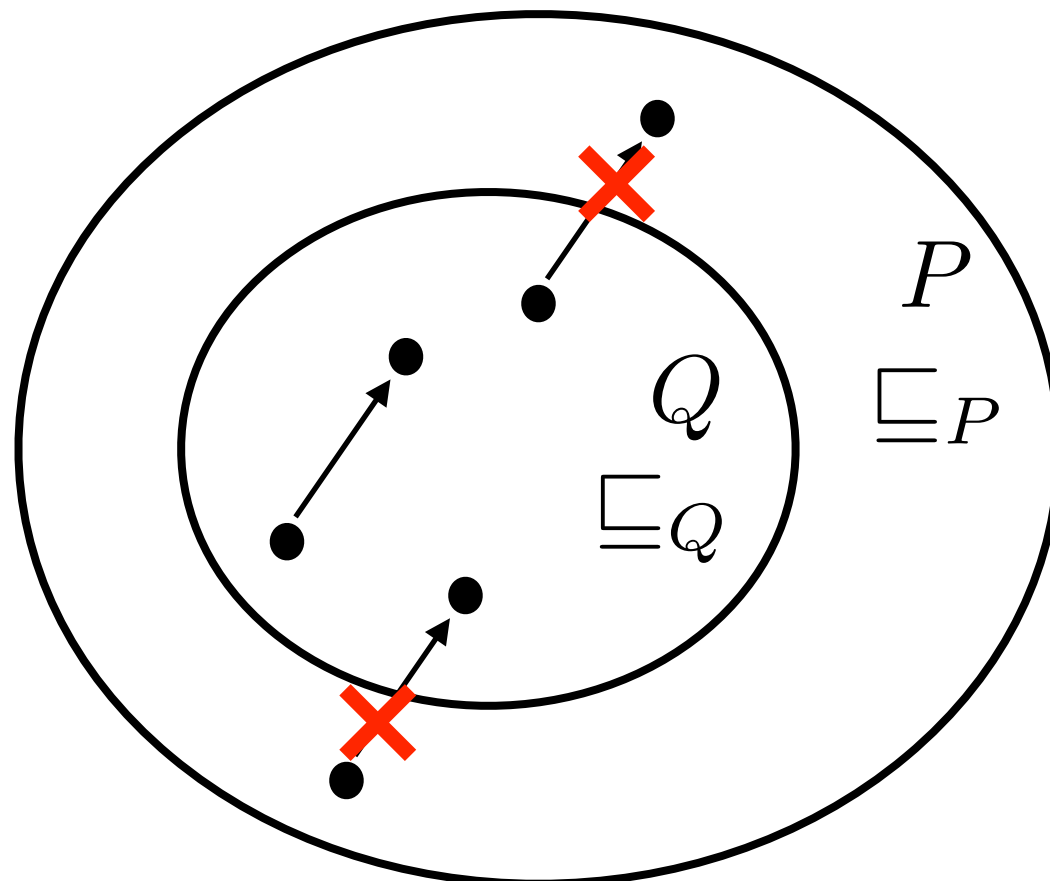
Subset of a PO

$(P, \sqsubseteq_P)$ PO $Q \subseteq P$

let $\sqsubseteq_Q \triangleq \sqsubseteq_P \cap (Q \times Q)$

TH. $(Q, \sqsubseteq_Q)$ is a PO

TH. if $(P, \sqsubseteq_P)$ is total, then $(Q, \sqsubseteq_Q)$ is total



PO $\sqsubseteq$

w.f. $\prec$

reflexive

not reflexive (otherwise cycle!)

antisymmetric

antisymmetric (otherwise cycle!)
 $p \prec q \wedge q \prec p$ is always false

transitive

can be transitive ($\prec^+$ w.f.)

has infinite descending chains
(if nonempty)

no infinite descending chain

$\prec^*$ is always a PO

$\sqsubseteq$ can be w.f.

Element properties (least, minimal, ...)

Least element

$(P, \sqsubseteq)$ PO $Q \subseteq P$ $\ell \in Q$

ℓ is a **least** element of Q if $\forall q \in Q. \ell \sqsubseteq q$

TH. (uniqueness of least element)

$(P, \sqsubseteq)$ PO $Q \subseteq P$ ℓ_1, ℓ_2 least elements of Q implies $\ell_1 = \ell_2$

$$\left. \begin{array}{l} \ell_1 \text{ least element of } Q \Rightarrow \ell_1 \sqsubseteq \ell_2 \\ \ell_2 \text{ least element of } Q \Rightarrow \ell_2 \sqsubseteq \ell_1 \end{array} \right\} \Rightarrow \ell_1 = \ell_2$$

by antisymmetry

Bottom

$(P, \sqsubseteq)$ PO the least element of P
(if it exists) is called **bottom** and denoted $\perp$

sometimes written $\perp_P$

Examples

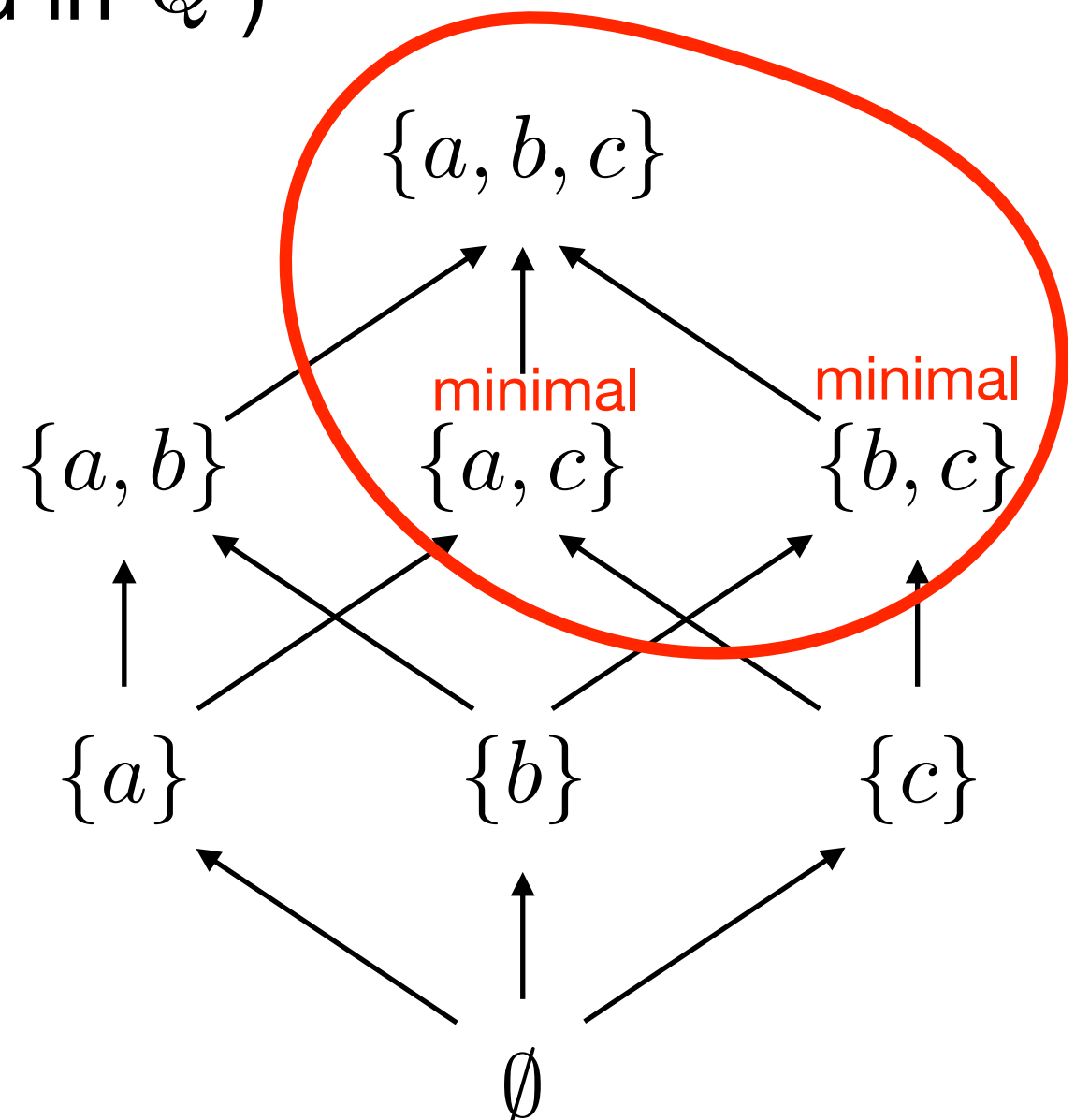
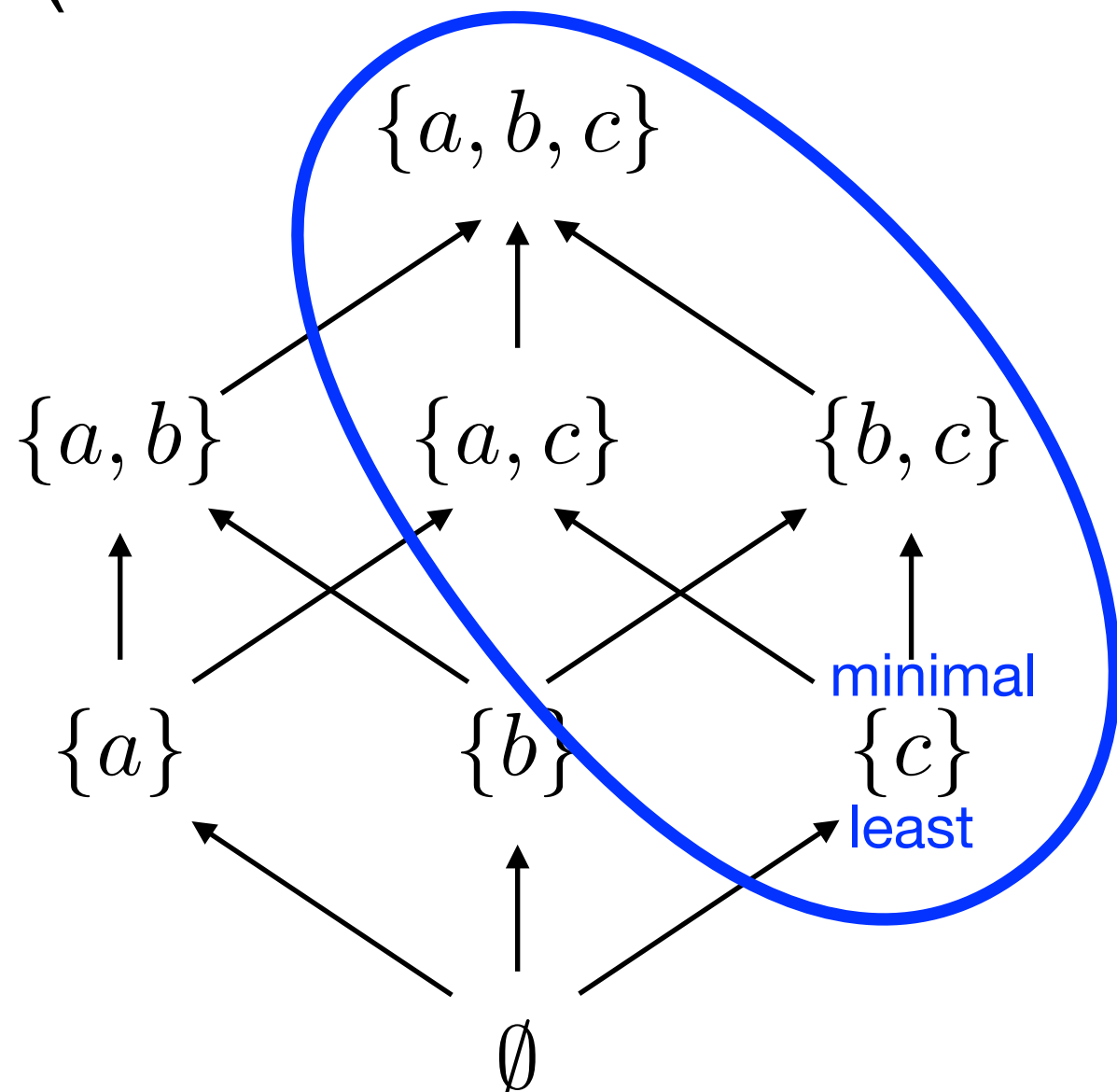
PO	bottom?
$(\mathbb{N} \cup \{\infty\}, \leq)$	0
$(\wp(S), \subseteq)$	$\emptyset$
$(\mathbb{Z}, \leq)$	
$(\mathbb{Z} \cup \{-\infty, \infty\}, \leq)$	$-\infty$

Minimal element

$(P, \sqsubseteq)$ PO $Q \subseteq P$ $m \in Q$

m is a **minimal** element of Q if $\forall q \in Q. \quad q \sqsubseteq m \Rightarrow q = m$

(no smaller element can be found in Q)



Least vs minimal

least $\forall q \in Q. \ell \sqsubseteq q$

minimal $\forall q \in Q. q \sqsubseteq m \Rightarrow q = m$

unique

not necessarily unique

minimal

not necessarily least
can be least

Reverse order

TH. $(P, \sqsubseteq)$ PO implies $(P, \sqsupseteq)$ PO

note:
 $\sqsupseteq = \sqsubseteq^{-1}$

proof. it is immediate to check that $\sqsupseteq$

- is reflexive
- is antisymmetric
- is transitive

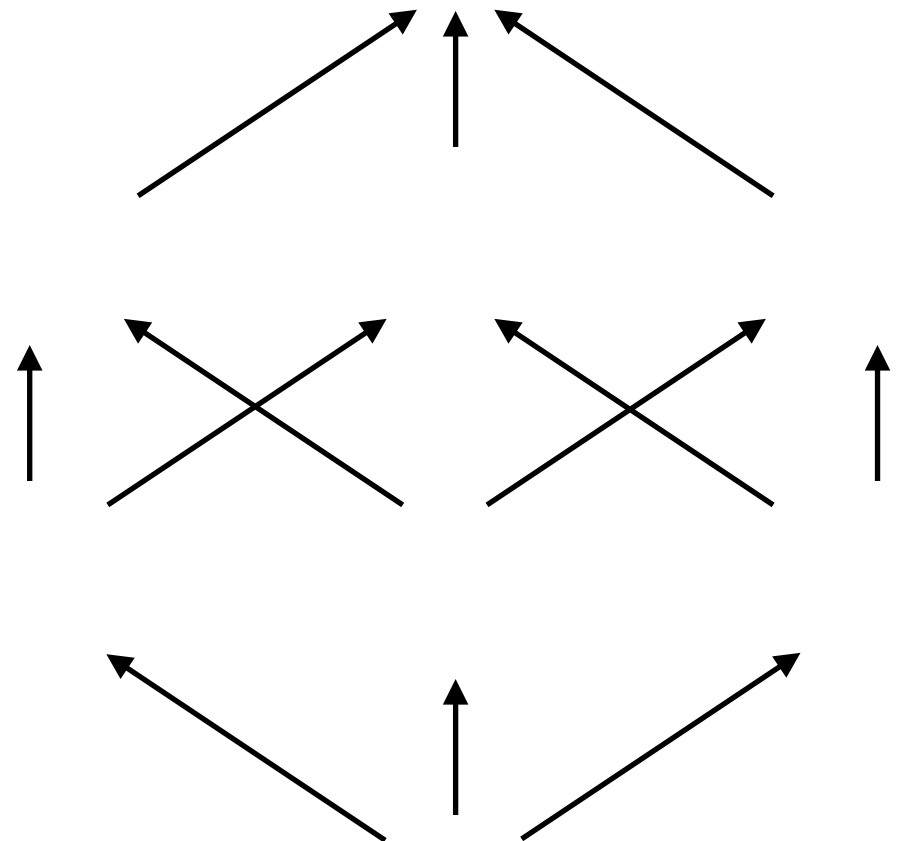
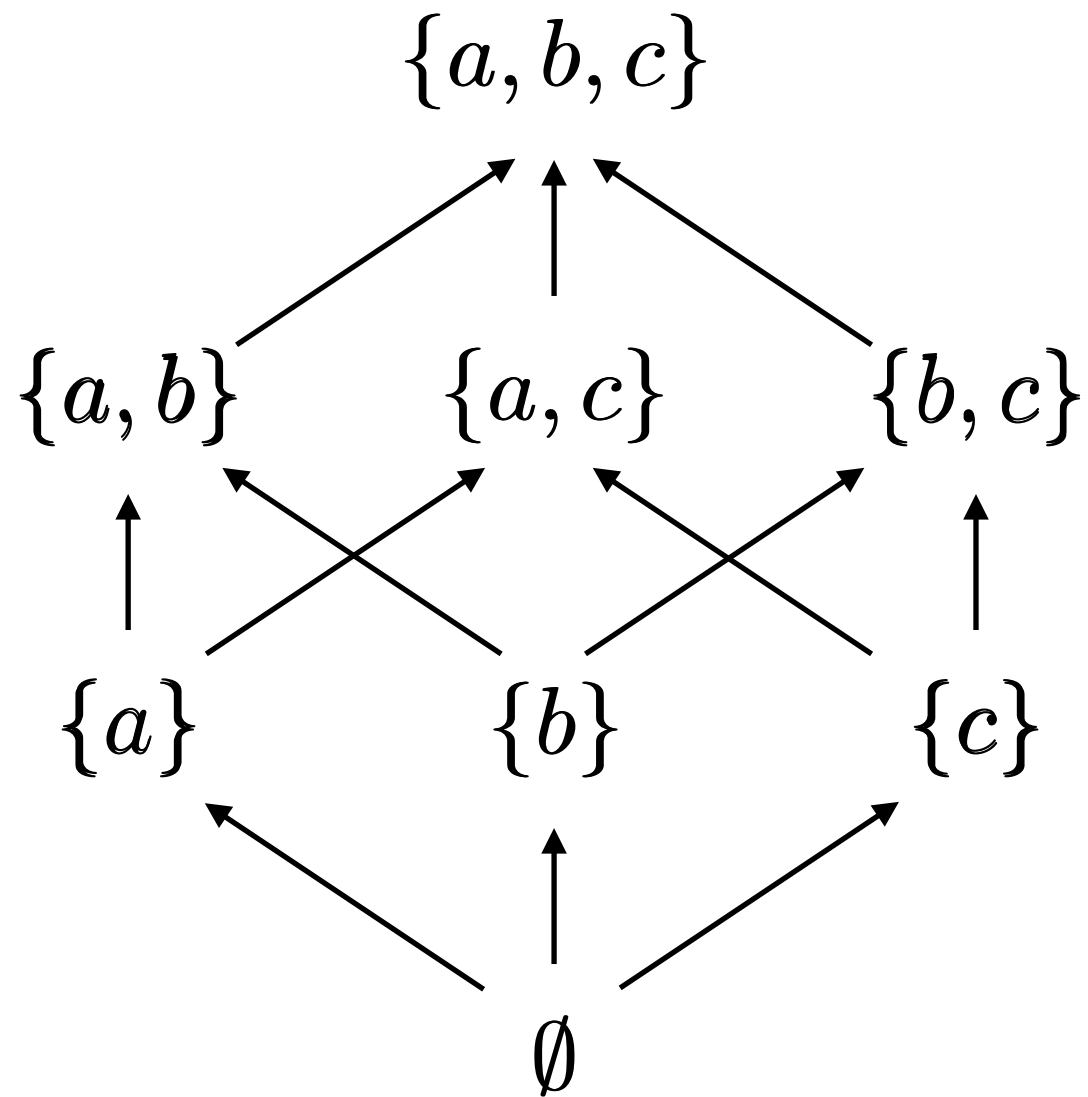
$(P, \sqsubseteq)$ PO $Q \subseteq P$

greatest element: least element of Q w.r.t. $(P, \sqsupseteq)$

top element: $\top$ greatest element of P (if it exists)

maximal element: minimal element of Q w.r.t. $(P, \sqsupseteq)$

Reversed powerset

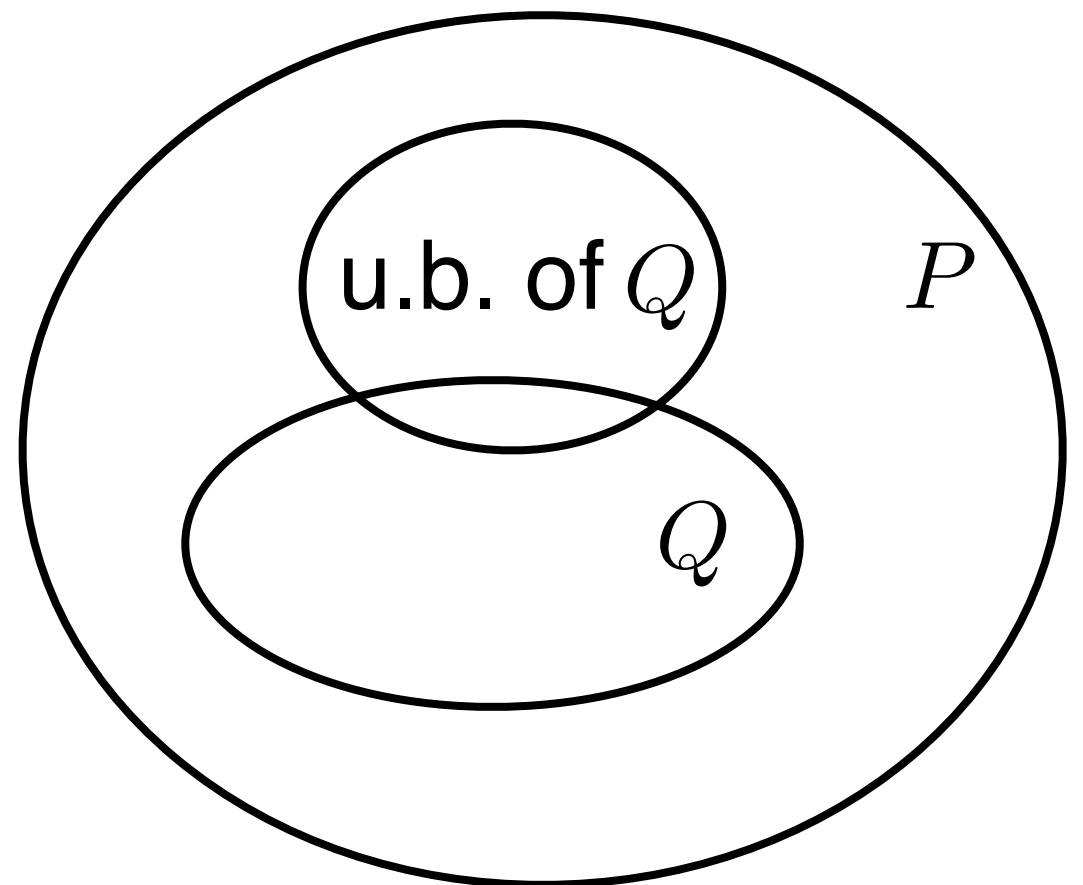
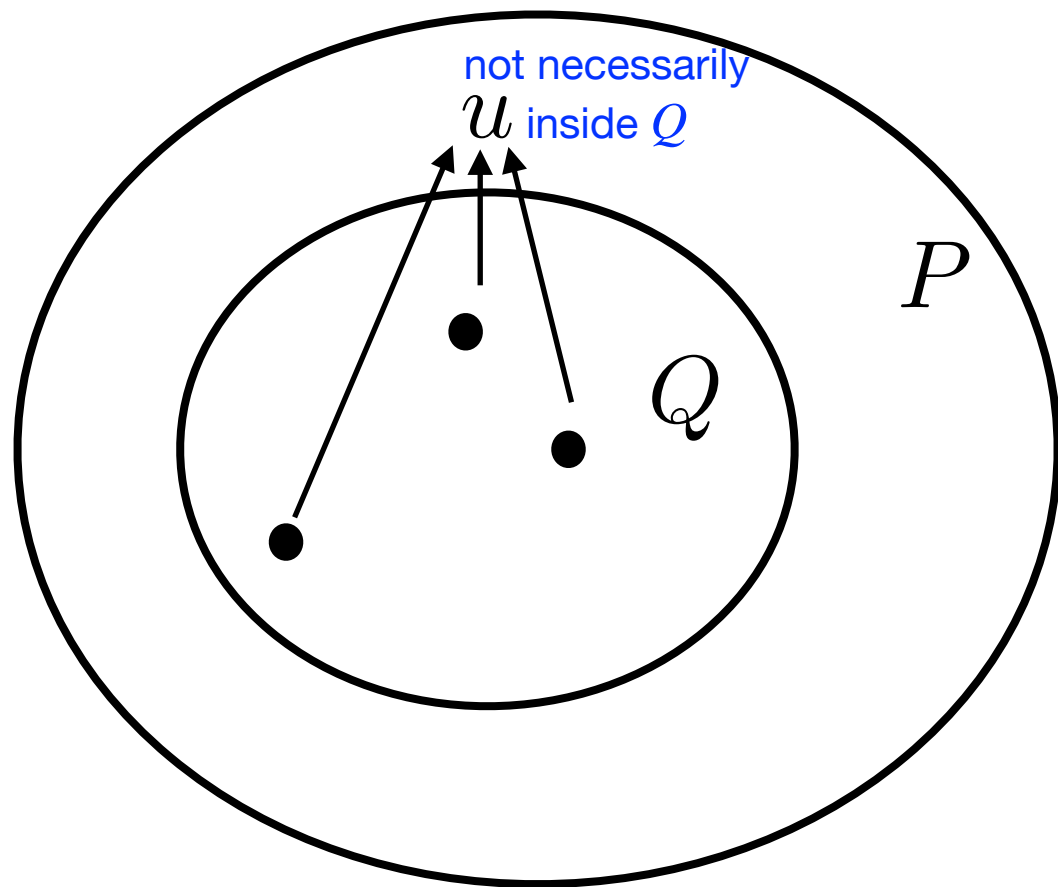


Upper bound

$(P, \sqsubseteq)$ PO $Q \subseteq P$ $u \in P$

u is an **upper bound** of Q if $\forall q \in Q. q \sqsubseteq u$
(all the elements of Q are smaller than u)

Q may have many upper bounds



Least upper bound

$(P, \sqsubseteq)$ PO $Q \subseteq P$ $p \in P$

p is the **least upper bound (lub)** of Q if

1. it is an upper bound of Q $\forall q \in Q. \quad q \sqsubseteq p$

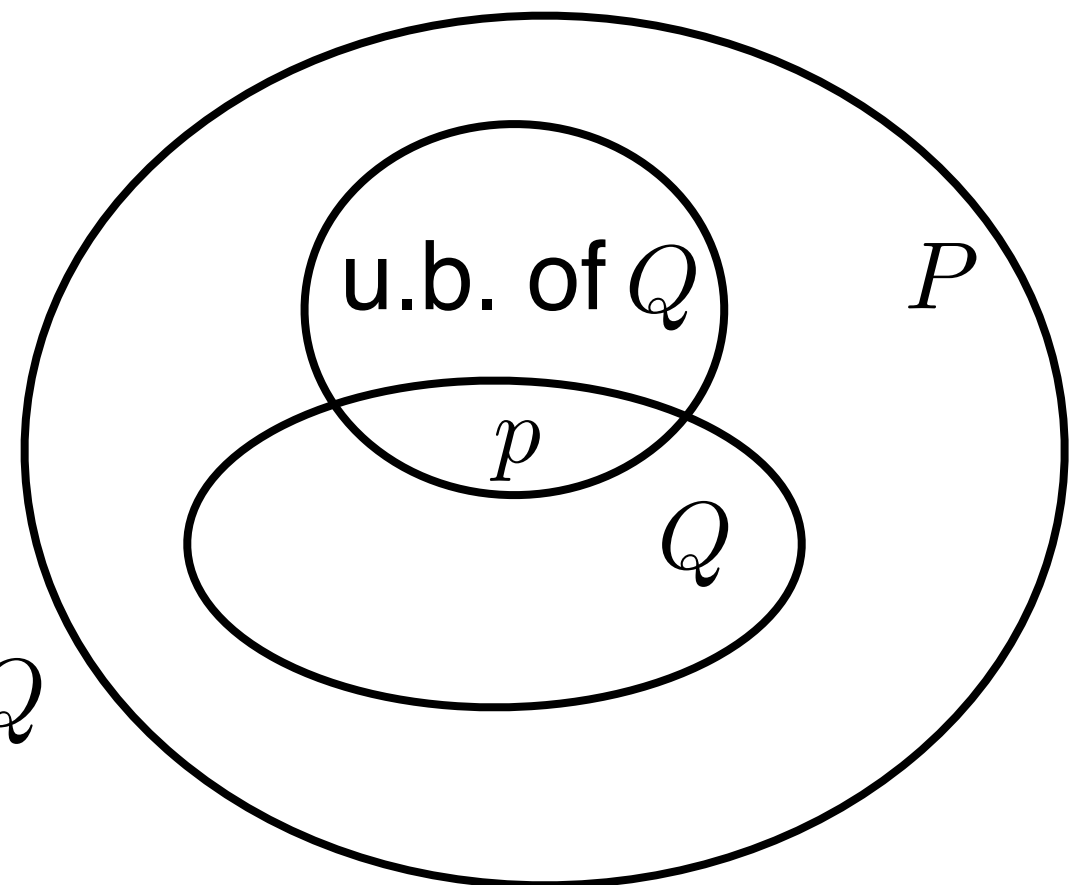
2. it is smaller than any other upper bound of Q

$$\forall u \in P. \quad (\forall q \in Q. q \sqsubseteq u) \Rightarrow p \sqsubseteq u$$

we write $p = \text{lub } Q$

intuitively, it is the least element that represents all of Q

p not necessarily an element of Q

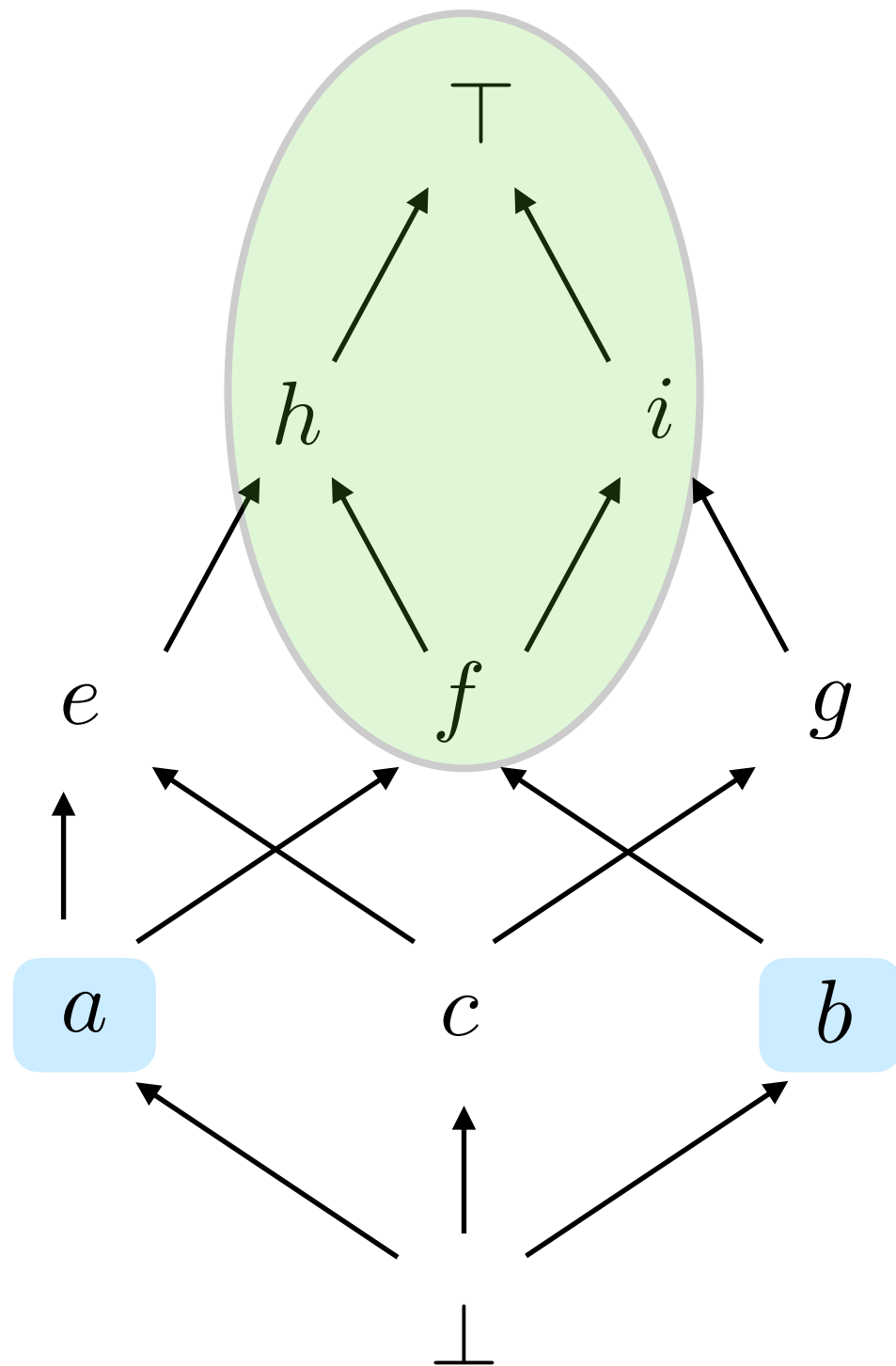




Exercise

Upper bounds of $\{a, b\}$? $\{f, h, i, \top\}$

lub? f

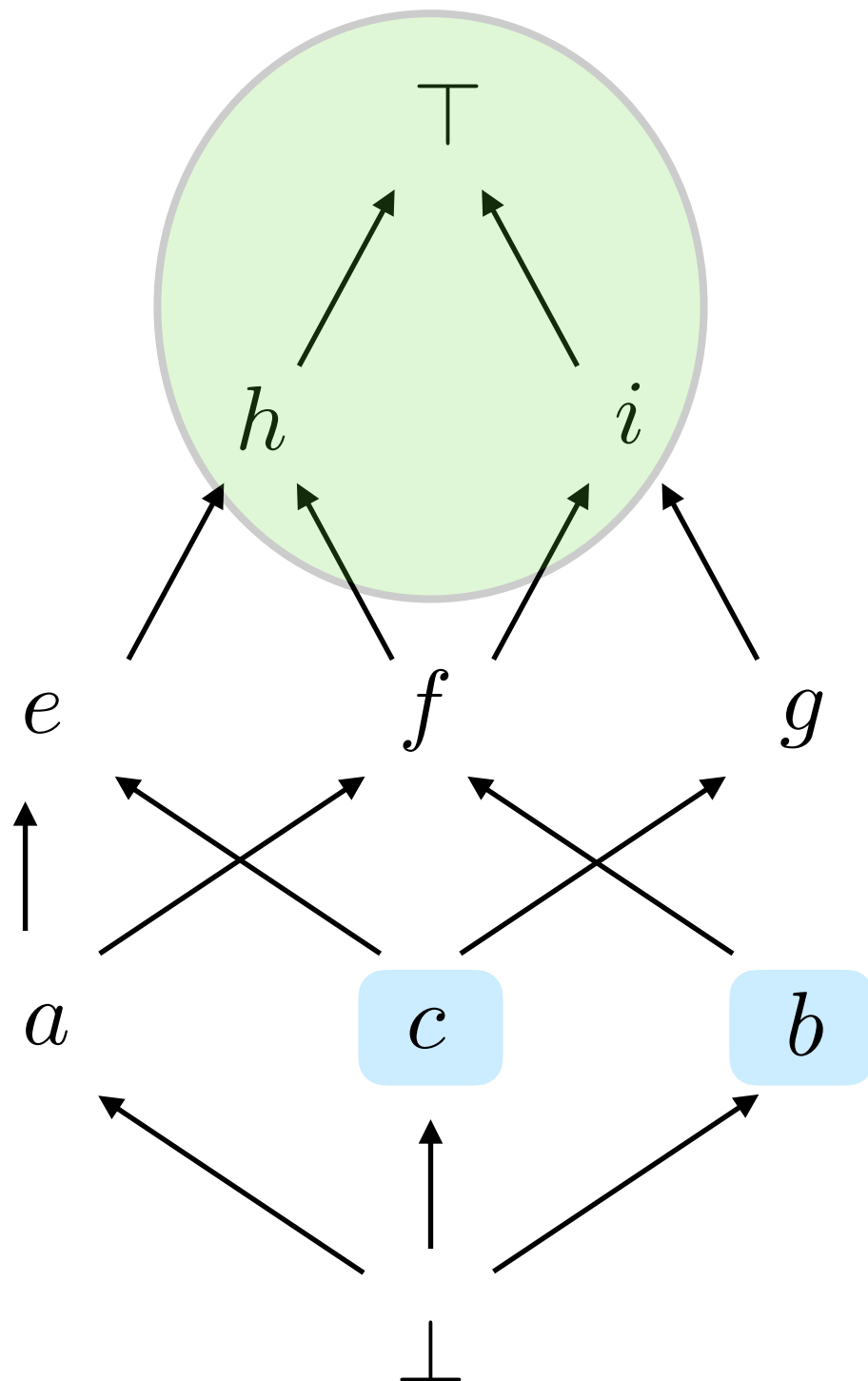




Exercise

Upper bounds of $\{b, c\}$? $\{h, i, \top\}$

lub? no lub!





Exercise

$(\mathbb{N}, \leq)$

$Q \subseteq \mathbb{N}$

lub?

if Q finite $\text{lub } Q = \max Q$
otherwise no lub

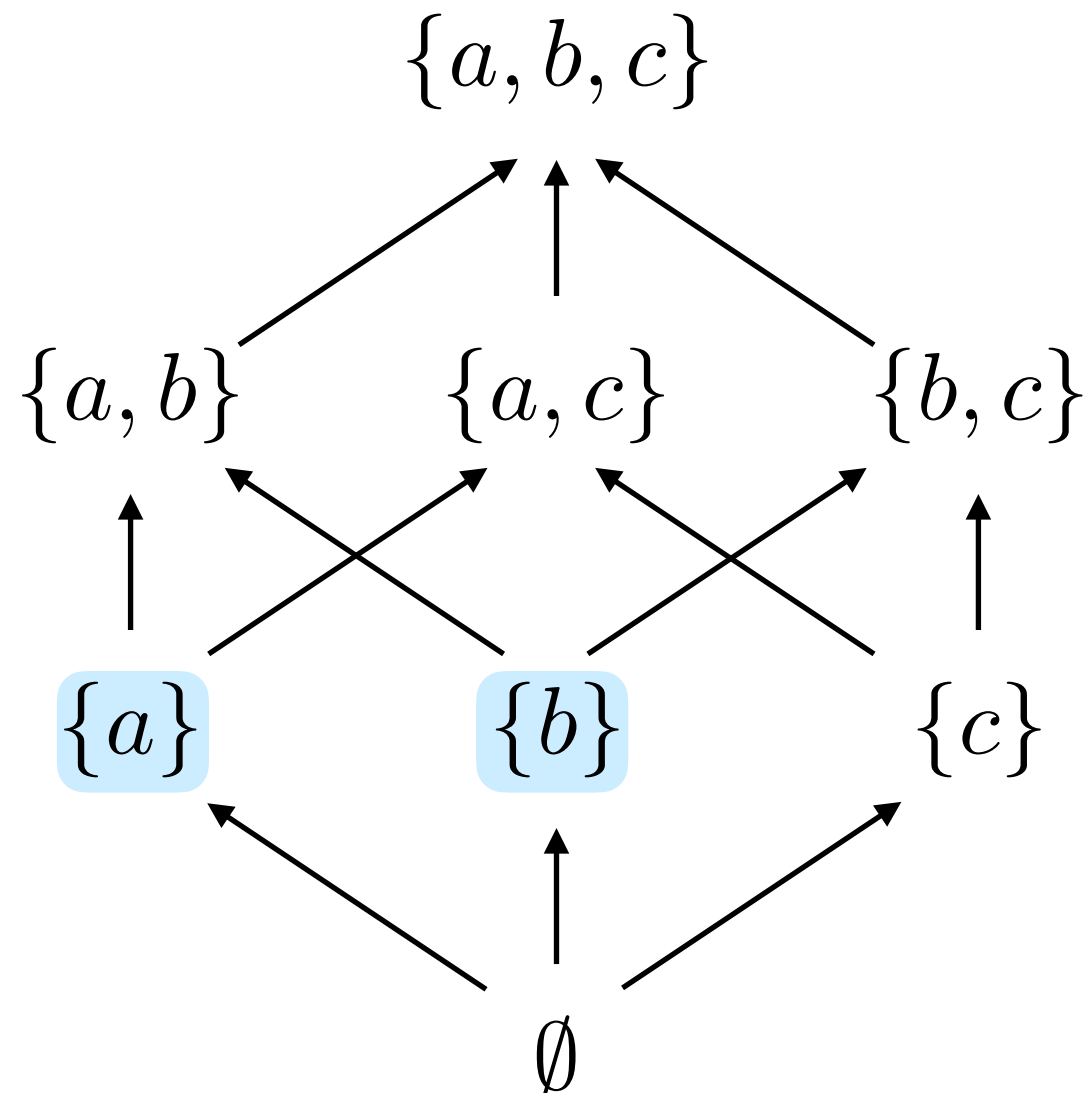




Exercise

$(\wp(S), \subseteq)$ $Q \subseteq \wp(S)$ lub?

$$\text{lub } Q = \bigcup_{T \in Q} T$$



$$\text{lub } \{\{a\}, \{b\}\} = \{a, b\}$$

Complete partial orders (CPO)

Completeness: the idea

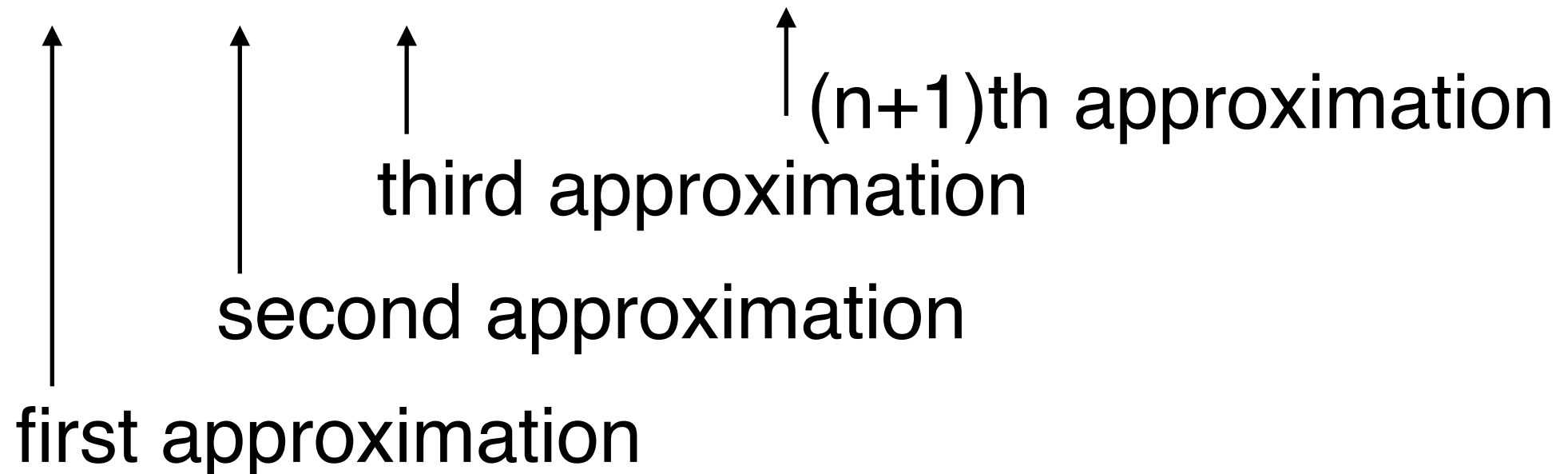
D a domain

$\sqsubseteq$ a way to compare element

$x \sqsubseteq y$ x is a (less precise) approximation of y
 x and y are consistent,
but y is more accurate than x

} PO

$$x_0 \sqsubseteq x_1 \sqsubseteq x_2 \sqsubseteq \cdots \sqsubseteq x_n \sqsubseteq \cdots$$



does any sequence of approximations tend to some limit?

Chain

$(P, \sqsubseteq)$ PO $\{d_i\}_{i \in \mathbb{N}}$ is a **chain** if $\forall i \in \mathbb{N}. d_i \sqsubseteq d_{i+1}$

$$d_0 \sqsubseteq d_1 \sqsubseteq d_2 \sqsubseteq \cdots \sqsubseteq d_n \sqsubseteq \cdots$$

any chain is an infinite list

finite chain: there are only finitely many distinct elements

$$\exists k \in \mathbb{N}. \forall i \geq k. d_i = d_{i+1}$$

or equivalently

$$\exists k \in \mathbb{N}. \forall i \geq k. d_i = d_k$$

Example

$(\mathbb{N}, \leq)$

$0 \leq 2 \leq 4 \leq \dots \leq 2n \leq \dots$ is an infinite chain

$0 \leq 1 \leq 3 \leq 3 \leq 5 \leq \dots \leq 5 \leq \dots$ is a finite chain

any chain has infinite length

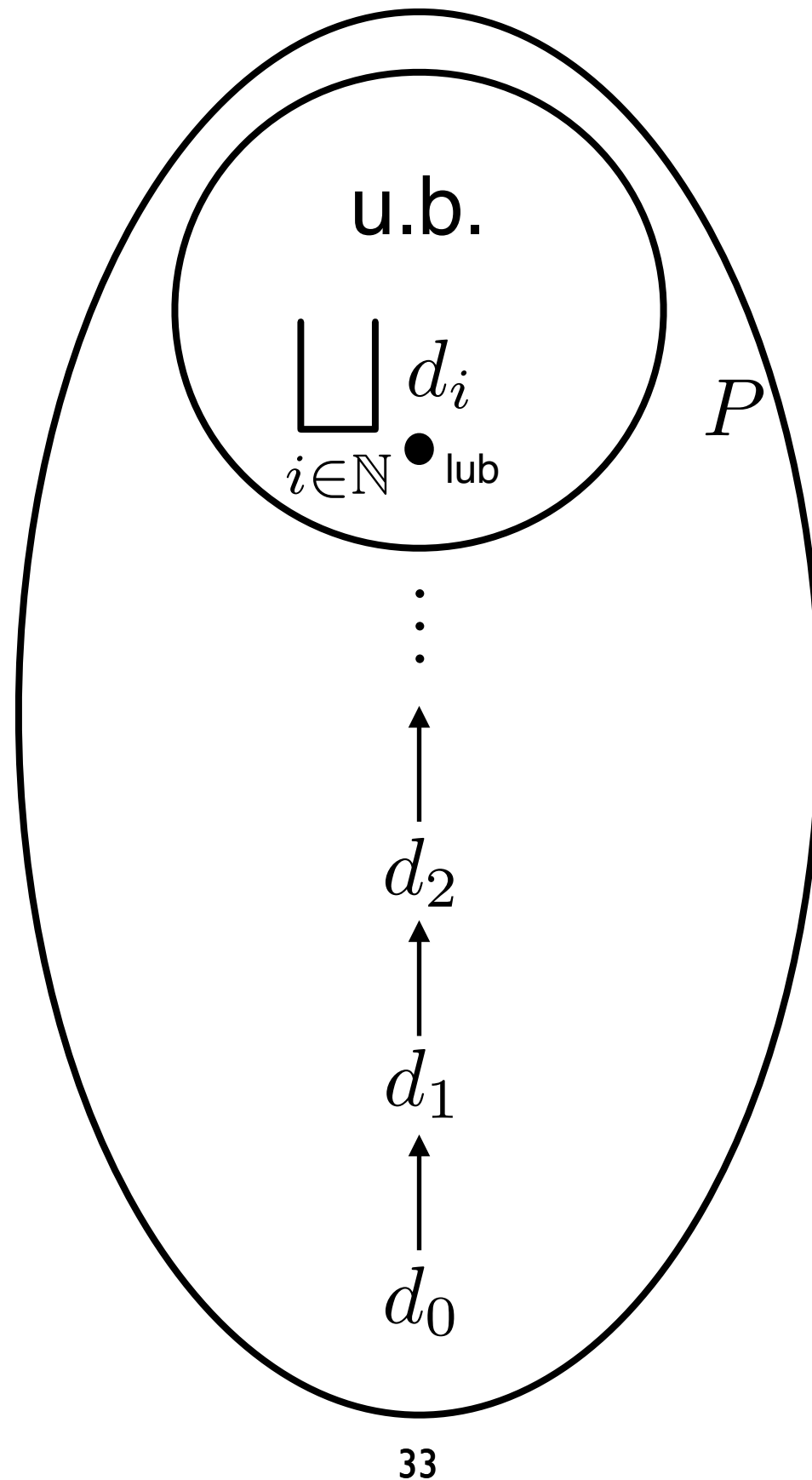
Limit of a chain

$(P, \sqsubseteq)$ PO $\{d_i\}_{i \in \mathbb{N}}$ a chain

we denote by $\bigsqcup_{i \in \mathbb{N}} d_i$ the lub of $\{d_i\}_{i \in \mathbb{N}}$ if it exists

and call it the **limit** of the chain

Limit illustrated



Example

$(\mathbb{N}, \leq)$

$0 \leq 2 \leq 4 \leq \dots \leq 2n \leq \dots$ has no lub
(empty set of upper bounds)

$0 \leq 1 \leq 3 \leq 3 \leq 5 \leq \dots \leq 5 \leq \dots$ has lub 5
(which upper bounds?)

Lemma on finite chains

Lemma (any finite chain has a limit)

$$(P, \sqsubseteq) \text{ PO} \quad \{d_i\}_{i \in \mathbb{N}} \text{ a finite chain} \quad \Rightarrow \quad \bigsqcup_{i \in \mathbb{N}} d_i \text{ exists}$$

proof.

$$\{d_i\}_{i \in \mathbb{N}} \text{ finite} \Rightarrow \exists k. \forall i. d_{i+k} = d_k$$

the elements of the chain are totally ordered

d_k is the greatest element of the chain

d_k is an upper bound $\forall i. d_i \sqsubseteq d_k$

d_k is the least upper bound

take u such that $\forall i. d_i \sqsubseteq u$ then $d_k \sqsubseteq u$

Prefix independence

Lemma (prefix independence) $(P, \sqsubseteq)$ PO $\{d_i\}_{i \in \mathbb{N}}$ a chain

$$\text{if } \bigsqcup_{i \in \mathbb{N}} d_i \text{ exists } \Rightarrow \forall k. \bigsqcup_{i \in \mathbb{N}} d_{i+k} = \bigsqcup_{i \in \mathbb{N}} d_i$$

$$\begin{array}{ccc} d_0 \sqsubseteq d_1 \sqsubseteq d_2 \sqsubseteq \cdots \sqsubseteq d_k \sqsubseteq d_{k+1} \sqsubseteq \cdots & \bigsqcup_{i \in \mathbb{N}} d_i & \\ & = & \\ & d_k \sqsubseteq d_{k+1} \sqsubseteq \cdots & \bigsqcup_{i \in \mathbb{N}} d_{i+k} \end{array}$$

Prefix independence

Lemma (prefix independence) $(P, \sqsubseteq)$ PO $\{d_i\}_{i \in \mathbb{N}}$ a chain

$$\text{if } \bigsqcup_{i \in \mathbb{N}} d_i \text{ exists } \Rightarrow \forall k. \bigsqcup_{i \in \mathbb{N}} d_{i+k} = \bigsqcup_{i \in \mathbb{N}} d_i$$

proof.

take a generic k

we prove that $\{d_i\}_{i \in \mathbb{N}}$ and $\{d_{i+k}\}_{i \in \mathbb{N}}$ have the same u.b.
(and thus the same lub)

1. if u is an u.b. of $\{d_i\}_{i \in \mathbb{N}}$ then is an u.b. of $\{d_{i+k}\}_{i \in \mathbb{N}}$

because $\{d_{i+k}\}_{i \in \mathbb{N}} \subseteq \{d_i\}_{i \in \mathbb{N}}$

2. if u is an u.b. of $\{d_{i+k}\}_{i \in \mathbb{N}}$ we need to show $\forall j. d_j \sqsubseteq u$
for $j \geq k$ it is obvious $\in \{d_i\}_{i \in \mathbb{N}}$

if $j < k$ then $d_j \sqsubseteq d_k \sqsubseteq u$ because $d_k \in \{d_{i+k}\}_{i \in \mathbb{N}}$

Complete partial order

$(P, \sqsubseteq)$ PO P is **complete** if each chain has a limit (lub)

TH. Any finite chain has a limit
(the last element in the sequence)

If P has only finite chains it is complete

If P is finite it is complete

Any discrete order is complete

Any flat order is complete

Example

$(\mathbb{N}, \leq)$ is not complete
(it is enough to exhibit a chain with no limit)

$0 \leq 2 \leq 4 \leq \dots \leq 2n \leq \dots$ has no lub
(empty set of u.b.)

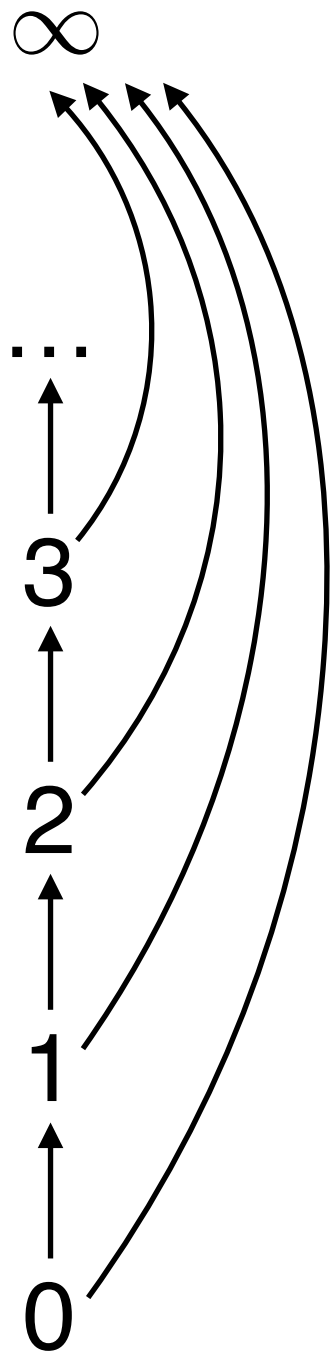


Exercise

$(\mathbb{N} \cup \{\infty\}, \leq)$

complete? 

any infinite chain has limit ∞
(set of u.b. $\{\infty\}$)





Exercise

$(\wp(S), \subseteq)$

complete? 

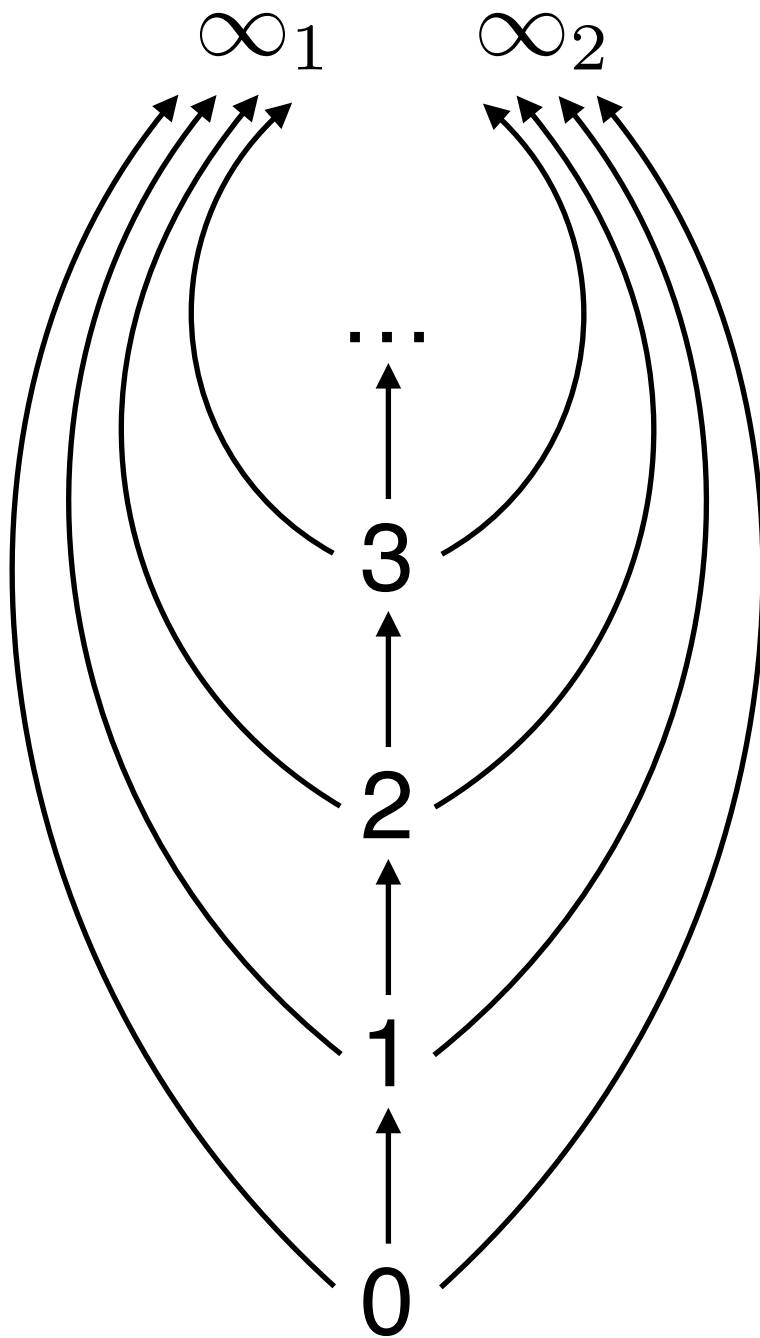
$$\{S_i\}_{i \in \mathbb{N}} \quad \bigsqcup_{i \in \mathbb{N}} S_i = \bigcup_{i \in \mathbb{N}} S_i = \{x \mid \exists k \in \mathbb{N}. x \in S_k\}$$

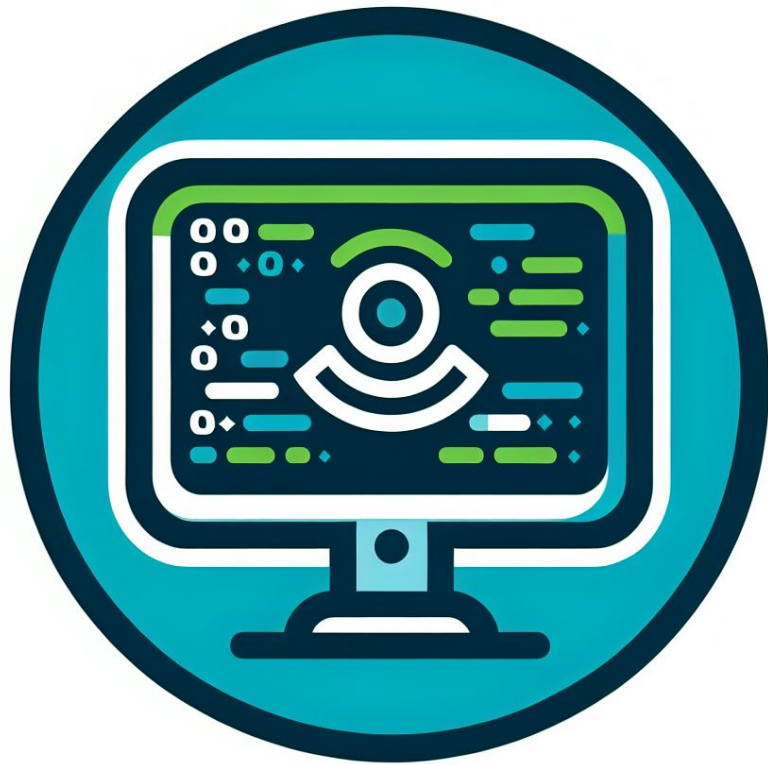


Exercise

$(\mathbb{N} \cup \{\infty_1, \infty_2\}, \leq)$ complete? 

any infinite chain has no limit
(set of u.b. $\{\infty_1, \infty_2\}$)





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08b - Kleene's fixed point theorem

Partial functions

Comparing functions

given two functions $f, g : A \rightarrow B$, when can we say $f = g$?

$$\forall a \in A . f(a) = g(a)$$

if we see functions as relations

$$\{ (a, f(a)) \mid a \in A \} \subseteq A \times B$$

we can use set equality

Example

$$f(n) = n!$$

$$f = \{ \begin{array}{l} \overset{n}{\quad} \overset{f(n)}{\quad} \\ (0, 1), \\ (1, 1), \\ (2, 2), \\ (3, 6), \\ \dots \\ (k, k!), \\ \dots \end{array} \}$$

$$f(n) = n(n+1)/2$$

$$f = \{ \begin{array}{l} (0, 0), \\ (1, 1), \\ (2, 3), \\ (3, 6), \\ \dots \\ (k, T_k), \\ \dots \end{array} \}$$

Partial functions

let $f: A \rightarrow B$, or equivalently $f: A \rightarrow B \cup \{ \perp \}$

the function f can be undefined on some inputs

we can still see partial functions as relations

$$\{ (a, f(a)) \mid a \in A, f(a) \neq \perp \} \subseteq A \times B$$

omit pairs where $f(a)$ is undefined

Partial functions

$D = (A \multimap B) = \mathbf{Pf}(A, B) = \{f : A \multimap B\}$ partial functions

$f \sqsubseteq g$ if $f(a)$ is defined, $g(a)$ is defined and $g(a) = f(a)$

but $g(a)$ can be defined when $f(a)$ is not

if we see partial functions as relations

$$\{(x, f(x)) \mid f(x) \neq \perp\} \subseteq A \times B$$

$f \sqsubseteq g$ means essentially $f \subseteq g$

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ even} \\ \perp & \text{otherwise} \end{cases}$$

$$f = \{ \begin{array}{c} \textcolor{red}{n} \quad \textcolor{red}{f(n)} \\ (0, 0), \\ (2, 1), \\ (4, 2), \\ (6, 3), \\ \dots \\ (2k, k), \\ \dots \end{array} \}$$

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$g(n) = \begin{cases} n/2 & \text{if } n \text{ even} \\ 2 \cdot n & \text{otherwise} \end{cases}$$

$$g = \{ \begin{array}{l} (0, 0), (1, 2), \\ (2, 1), (3, 6), \\ (4, 2), (5, 10), \\ (6, 3), (7, 14), \\ \dots \\ (2k, k), (1 + 2k, 2 + 4k), \\ \dots \end{array} \}$$

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$g(n) = \begin{cases} n/2 & \text{if } n \text{ even} \\ 2 \cdot n & \text{otherwise} \end{cases}$$

$$g = \{ \begin{array}{l} (0, 0), (1, 2), \\ (2, 1), (3, 6), \\ (4, 2), (5, 10), \\ (6, 3), (7, 14), \\ \dots \\ (2k, k), (1 + 2k, 2 + 4k), \\ \dots \end{array} \}$$

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ even} \\ \perp & \text{otherwise} \end{cases}$$

$$f = \{ \begin{array}{l} (0, 0), \\ (2, 1), \\ (4, 2), \\ (6, 3), \\ \dots \\ (2k, k), \\ \dots \end{array} \}$$

$$\begin{array}{ll} f \sqsubseteq g? & \checkmark \\ g \sqsubseteq f? & \times \end{array}$$

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \sqsubseteq \{ (0,0) \} \sqsubseteq \{ (0,0), \quad \sqsubseteq \dots \\ (1,1) \}$$

which function(s) are we approximating?

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \subseteq \{ (0,0) \} \subseteq \{ (0,0), (1,1) \} \subseteq \{ (0,0), (1,1), (2,2) \} \subseteq \dots$$

which function(s) are we approximating?

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \sqsubseteq \{ (0,0) \} \sqsubseteq \{ (0,0), (1,1) \} \sqsubseteq \{ (0,0), (1,1), (2,2) \} \sqsubseteq \{ (0,0), (1,1), (2,2), (3,3) \} \sqsubseteq \dots$$

which function(s) are we approximating?

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \sqsubseteq \{ (0,0) \} \sqsubseteq \{ (0,0), (1,1) \} \sqsubseteq \{ (0,0), (1,1), (2,2) \} \sqsubseteq \{ (0,0), (1,1), (2,2), (3,3) \} \sqsubseteq \{ (0,0), (1,1), (2,2), (3,3), (4,4) \} \sqsubseteq \dots$$

which function(s) are we approximating?

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \sqsubseteq \{ (0,1) \} \sqsubseteq \{ (0,1), (1,1) \} \sqsubseteq \{ (0,1), (1,1), (2,2) \} \sqsubseteq \dots$$

which function(s) are we approximating?

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \sqsubseteq \{ (0,1) \} \sqsubseteq \{ (0,1), (1,1) \} \sqsubseteq \{ (0,1), (1,1), (2,2) \} \sqsubseteq \{ (0,1), (1,1), (2,2), (3,6) \} \sqsubseteq \dots$$

which function(s) are we approximating?

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \sqsubseteq \{ (0,1) \} \sqsubseteq \{ (0,1), (1,1) \} \sqsubseteq \{ (0,1), (1,1), (2,2) \} \sqsubseteq \{ (0,1), (1,1), (2,2), (3,6) \} \sqsubseteq \{ (0,1), (1,1), (2,2), (3,6), (4,24) \} \sqsubseteq \dots$$

which function(s) are we approximating?

$n!$

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \sqsubseteq \{ (0,1) \} \sqsubseteq \{ (0,1), (4,24) \} \sqsubseteq \{ (0,1), (1,1), (4,24) \} \sqsubseteq \{ (0,1), (1,1), (3,6), (4,24) \} \sqsubseteq \{ (0,1), (1,1), (2,2), (3,6), (4,24) \} \sqsubseteq \dots$$

which function(s) are we approximating?

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \sqsubseteq \{ (1,1) \} \sqsubseteq \{ (1,1), (2,4) \} \sqsubseteq \{ (1,1), (2,4), (3,81) \} \sqsubseteq \{ (1,1), (2,4), (3,81), (4,256) \} \sqsubseteq \dots$$

which function(s) are we approximating?

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \sqsubseteq \{ (4,2) \} \sqsubseteq \{ (4,2), (6,3) \} \sqsubseteq \{ (4,2), (6,3), (8,4) \} \sqsubseteq \{ (4,2), (6,3), (8,4), (9,3) \} \sqsubseteq \{ (4,2), (6,3), (8,4), (9,3), (10,5) \} \sqsubseteq \dots$$

which function(s) are we approximating?

Example

$\mathbf{Pf}(\mathbb{N}, \mathbb{N})$

$$\emptyset \sqsubseteq \{ (1,6) \} \sqsubseteq \{ (1,6), (2,28) \} \sqsubseteq \{ (1,6), (2,28), (3,496) \} \sqsubseteq \{ (1,6), (2,28), (3,496), (4,8128) \} \sqsubseteq \dots$$

which function(s) are we approximating?

Functional property

$\mathbf{Pf}(A, B) = \{f : A \rightharpoonup B\}$ partial functions

$\mathbf{Pf}(A, B) = \{f \subseteq A \times B \mid \forall a \in A. \forall b_1, b_2 \in B. (a, b_1) \in f \wedge (a, b_2) \in f \Rightarrow b_1 = b_2\}$
functional property

Converge

$f(a) \downarrow \triangleq \exists b \in B. (a, b) \in f$ function f is defined on a

$f \sqsubseteq g \Leftrightarrow (\forall a \in A. f(a) \downarrow \Rightarrow (g(a) \downarrow \wedge f(a) = g(a)))$
 $\Leftrightarrow f \subseteq g$

$(\mathbf{Pf}(A, B), \sqsubseteq)$ is a PO with bottom

what is bottom?

is it complete?

the empty relation
(the function always undefined)

Is Pf complete?

$(\mathbf{Pf}(A, B), \sqsubseteq)$

complete?

Given a chain $\{f_i\}_{i \in \mathbb{N}}$ let us consider $\bigcup_{i \in \mathbb{N}} f_i \subseteq A \times B$

we want to prove that $\bigcup_{i \in \mathbb{N}} f_i \in \mathbf{Pf}(A, B)$

i.e. that $f = \bigcup_{i \in \mathbb{N}} f_i$ satisfies the functional property

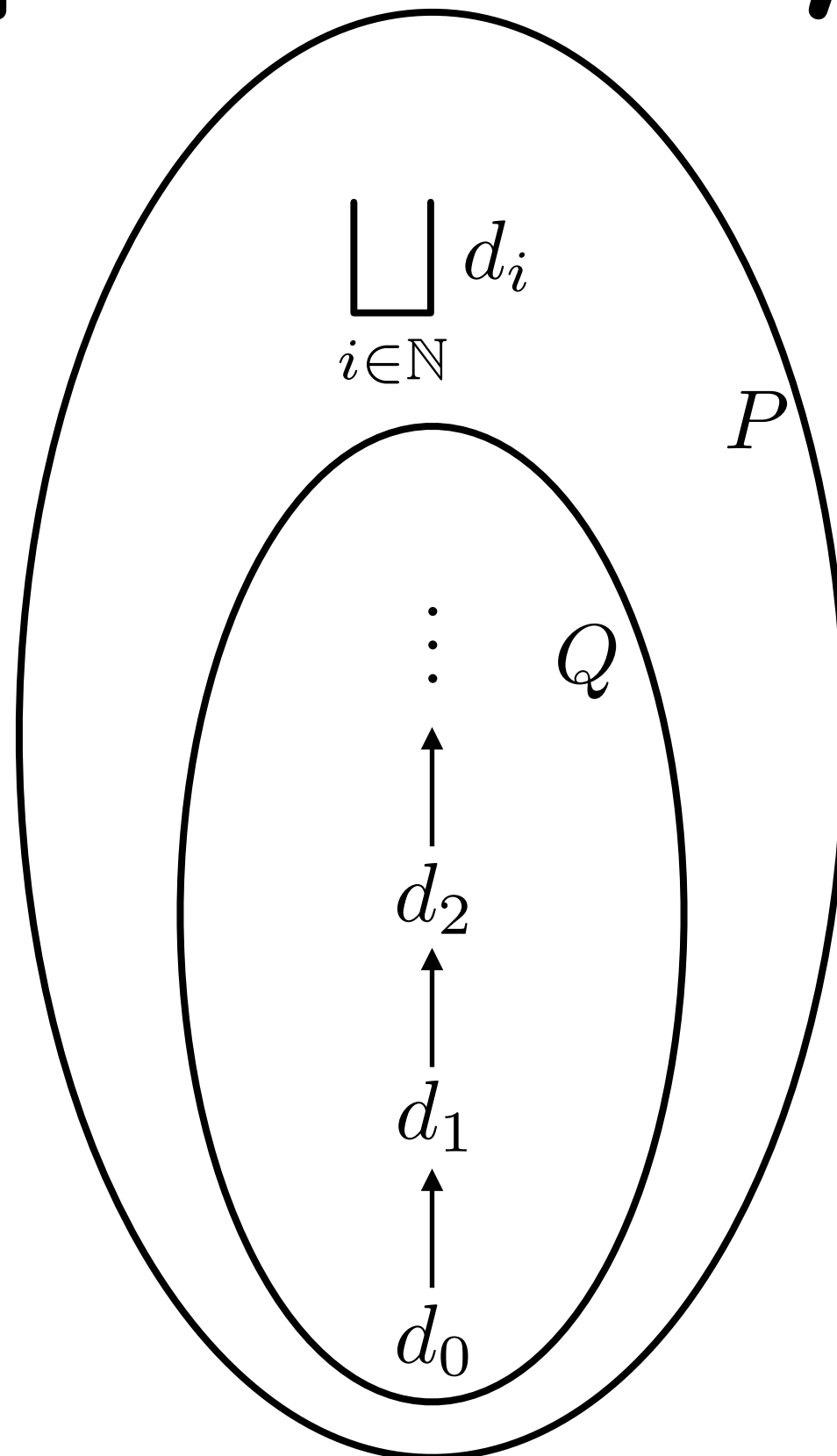
we know that each f_i is functional

$$\forall i \in \mathbb{N}. \forall a \in A. \forall b_1, b_2 \in B. (a, b_1) \in f_i \wedge (a, b_2) \in f_i \Rightarrow b_1 = b_2$$

we need to prove f is functional

$$\forall a \in A. \forall b_1, b_2 \in B. (a, b_1) \in f \wedge (a, b_2) \in f \Rightarrow b_1 = b_2$$

pictorially



is the limit in Q ?

Pf is complete

we need to prove f is functional

$$\forall a \in A. \forall b_1, b_2 \in B. (a, b_1) \in f \wedge (a, b_2) \in f \Rightarrow b_1 = b_2$$

Take $a \in A, b_1, b_2 \in B$ such that $(a, b_1) \in f \wedge (a, b_2) \in f$

we need to prove $b_1 = b_2$

$$(a, b_1) \in f = \bigcup_{i \in \mathbb{N}} f_i \Leftrightarrow \exists k \in \mathbb{N}. (a, b_1) \in f_k$$

$$m \triangleq \max\{k, h\}$$

$$(a, b_2) \in f = \bigcup_{i \in \mathbb{N}} f_i \Leftrightarrow \exists h \in \mathbb{N}. (a, b_2) \in f_h$$

Clearly $f_k \subseteq f_m$ $f_h \subseteq f_m$ f_m is functional

$$(a, b_1) \in f_m \quad (a, b_2) \in f_m \quad \Rightarrow \quad b_1 = b_2$$

Example

$$\begin{array}{lcl}
 \mathbf{Pf}(\mathbb{N}, \mathbb{N}) & f_0 & \emptyset \\
 & \subseteq & \{(0, 1)\} \\
 & \subseteq & \{(0, 1), (1, 1)\} \\
 & \subseteq & \{(0, 1), (1, 1), (2, 2)\} \\
 & \subseteq & \{(0, 1), (1, 1), (2, 2), (3, 6)\} \\
 & \subseteq & \{(0, 1), (1, 1), (2, 2), (3, 6), (4, 24)\} \\
 & \subseteq & \dots
 \end{array}$$

$\bigcup_{i \in \mathbb{N}} f_i$ is (maybe) the factorial function

note: the limit of partial functions can be a total function

Total functions

$\mathbf{Tf}(A, B) = (A \rightarrow B)$ total functions

$\mathbf{Pf}(A, B) \equiv \mathbf{Tf}(A, B_{\perp})$ $B_{\perp} \triangleq B \uplus \{\perp\}$
 $\sqsubseteq_{B_{\perp}} \triangleq$ flat order

$$f \sqsubseteq g \Leftrightarrow \forall a \in A. f(a) \sqsubseteq_{B_{\perp}} g(a)$$

PO? immediate to check

bottom? $f_{\perp}(a) = \perp$ for any $a \in A$

complete? we will prove it later

(as an instance of a more general result)

$$\left(\bigsqcup_{i \in \mathbb{N}} f_i\right)(a) \triangleq \bigsqcup_{i \in \mathbb{N}} f_i(a) \quad (\text{flat order, limit exists})$$

Monotone functions

Monotone function

$(D, \sqsubseteq_D)$ PO $(E, \sqsubseteq_E)$ PO $f : D \rightarrow E$

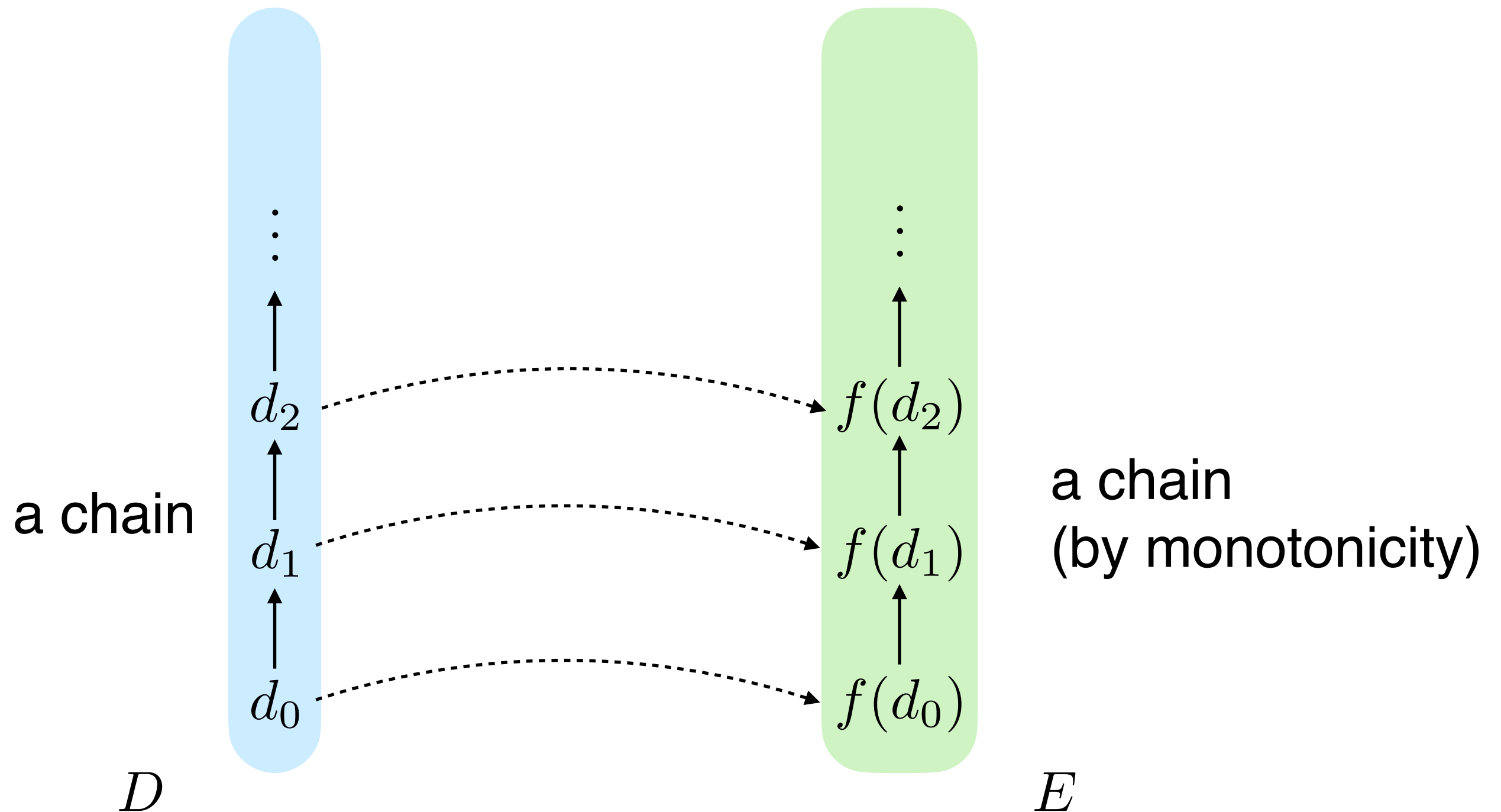
f is **monotone** if $\forall d_1, d_2 \in D. d_1 \sqsubseteq_D d_2 \Rightarrow f(d_1) \sqsubseteq_E f(d_2)$

Monotone = Order preserving

$\left. \begin{array}{l} \{d_i\}_{i \in \mathbb{N}} \text{ a chain in } D \\ f \text{ monotone} \end{array} \right\} \Rightarrow \{f(d_i)\}_{i \in \mathbb{N}} \text{ a chain in } E$

When $D = E$ we say $f : D \rightarrow D$ is a function on D

Monotonicity illustrated

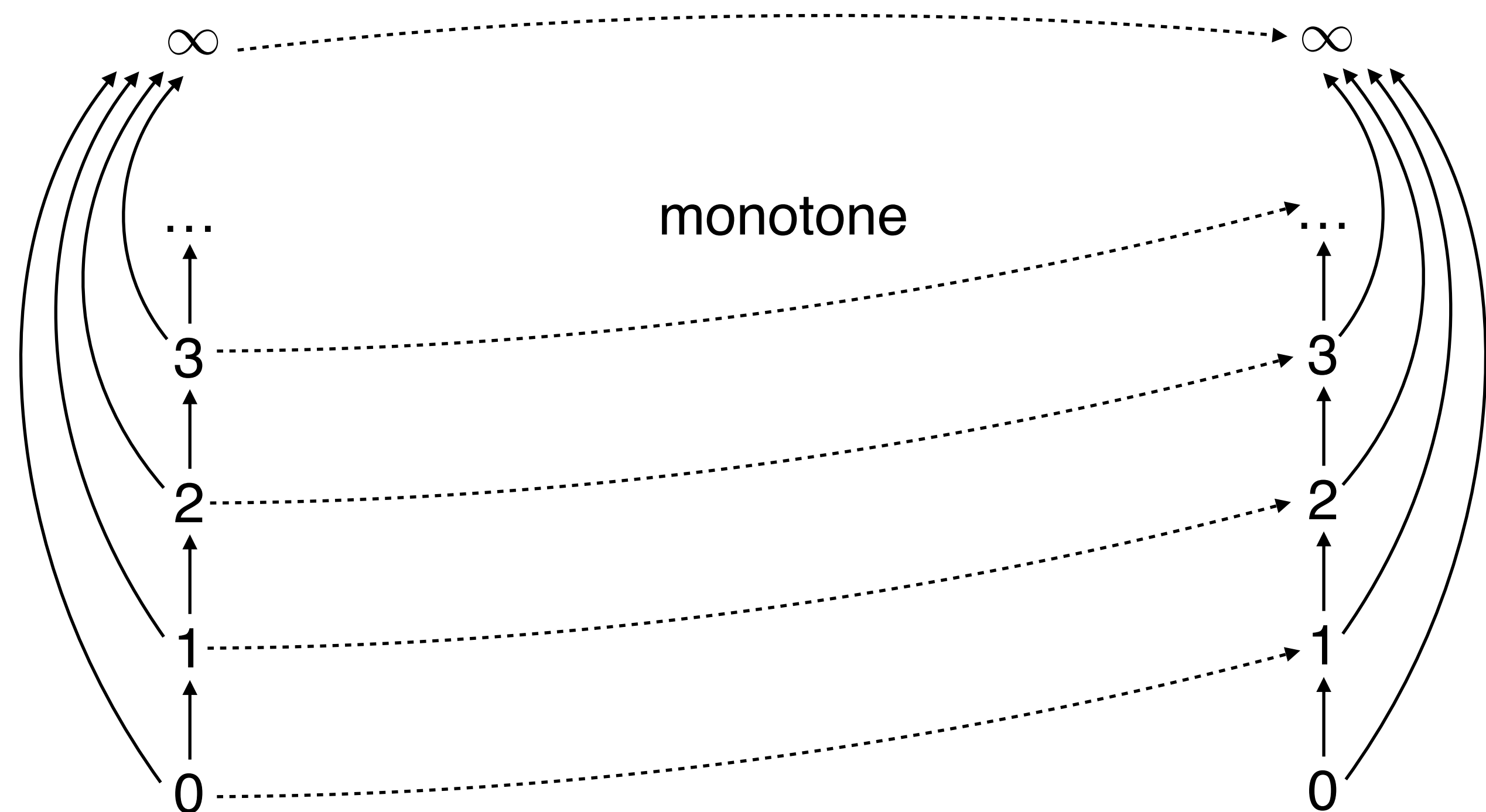


Example

$(\mathbb{N} \cup \{\infty\}, \leq)$

$$f(n) = n + 1$$
$$f(\infty) = \infty$$

$(\mathbb{N} \cup \{\infty\}, \leq)$





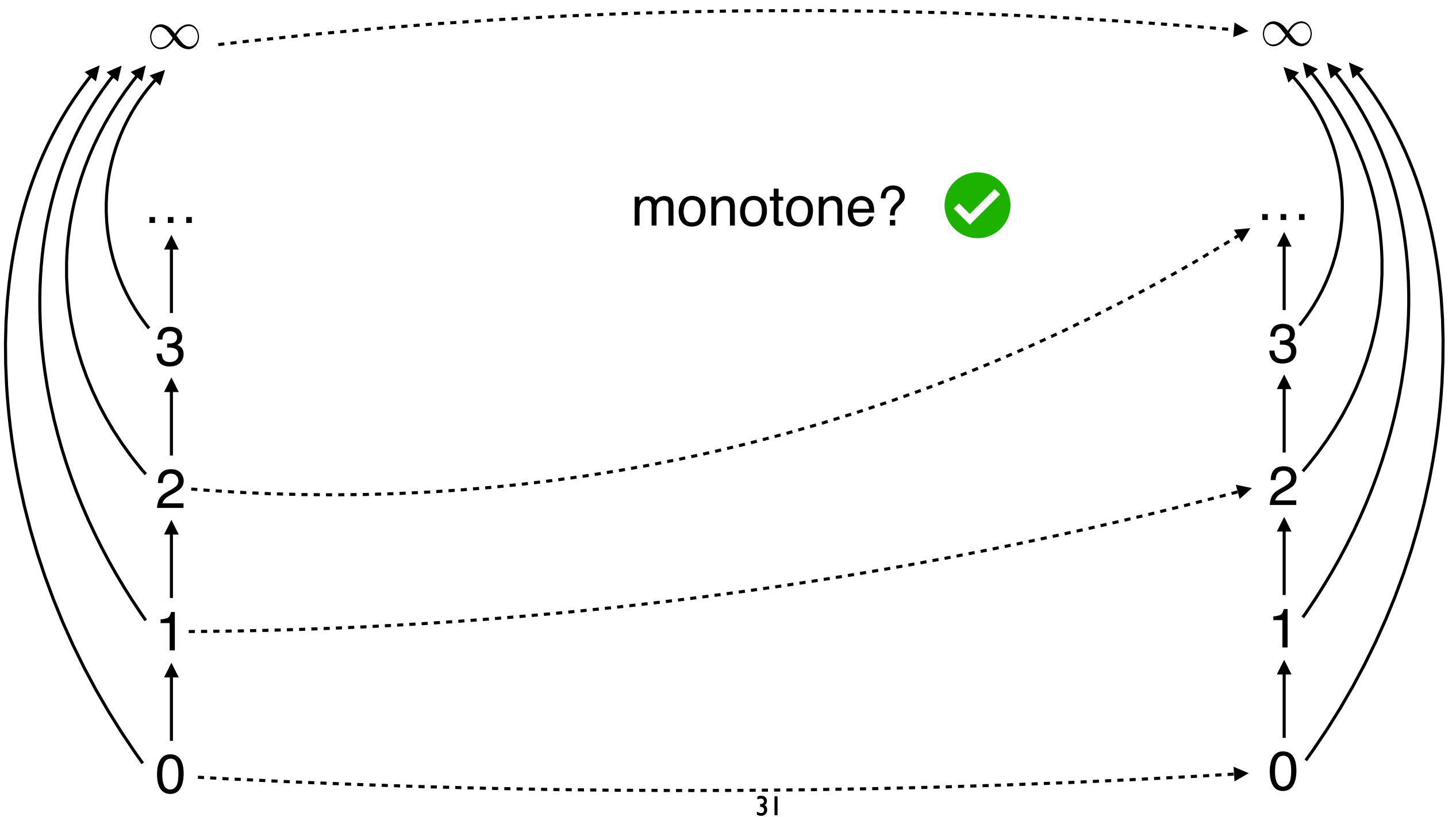
Exercise

$(\mathbb{N} \cup \{\infty\}, \leq)$

$$f(n) = 2 \cdot n$$
$$f(\infty) = \infty$$

$(\mathbb{N} \cup \{\infty\}, \leq)$

monotone?





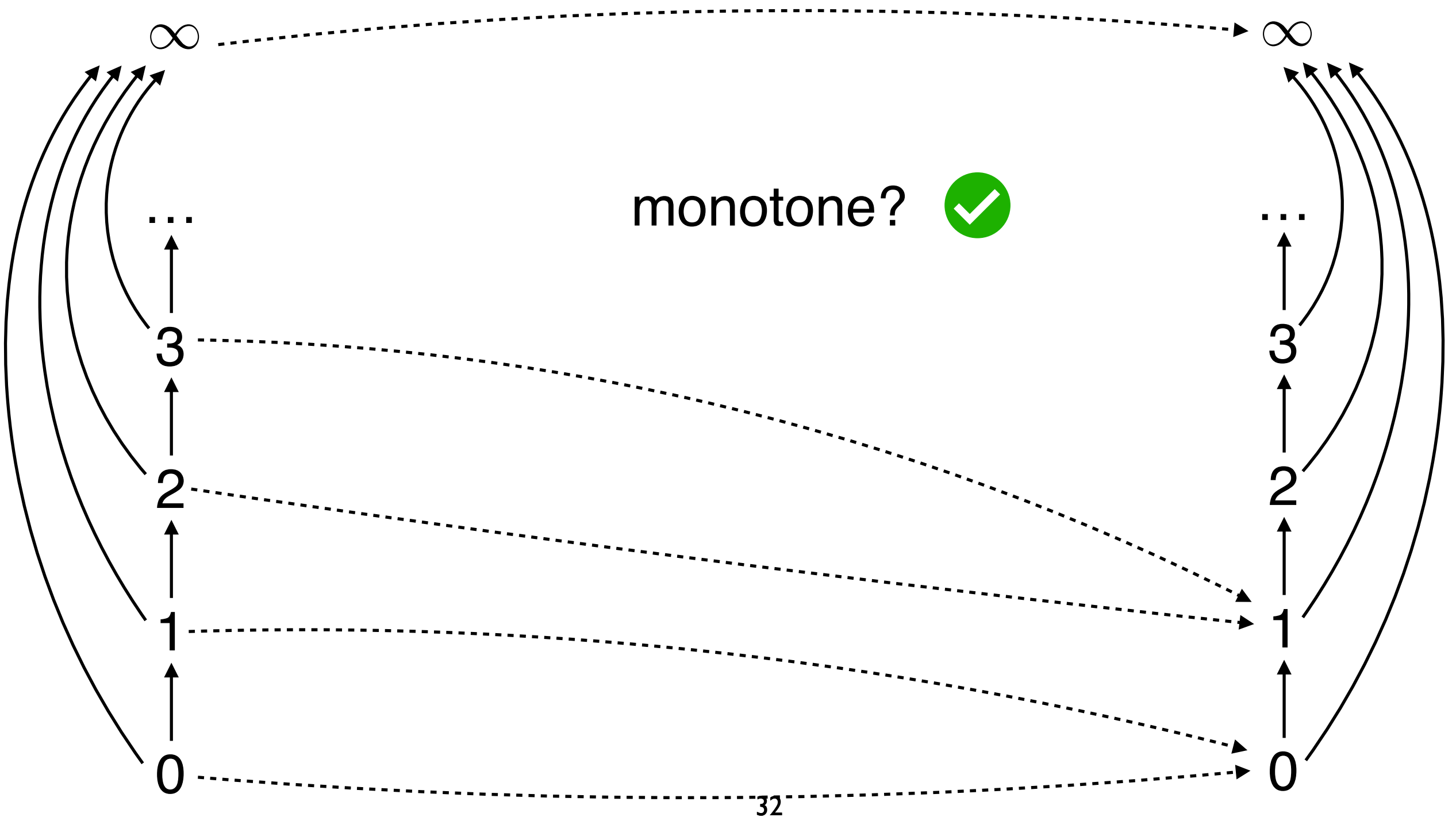
Exercise

$(\mathbb{N} \cup \{\infty\}, \leq)$

$$f(n) = n/2$$
$$f(\infty) = \infty$$

$(\mathbb{N} \cup \{\infty\}, \leq)$

monotone?





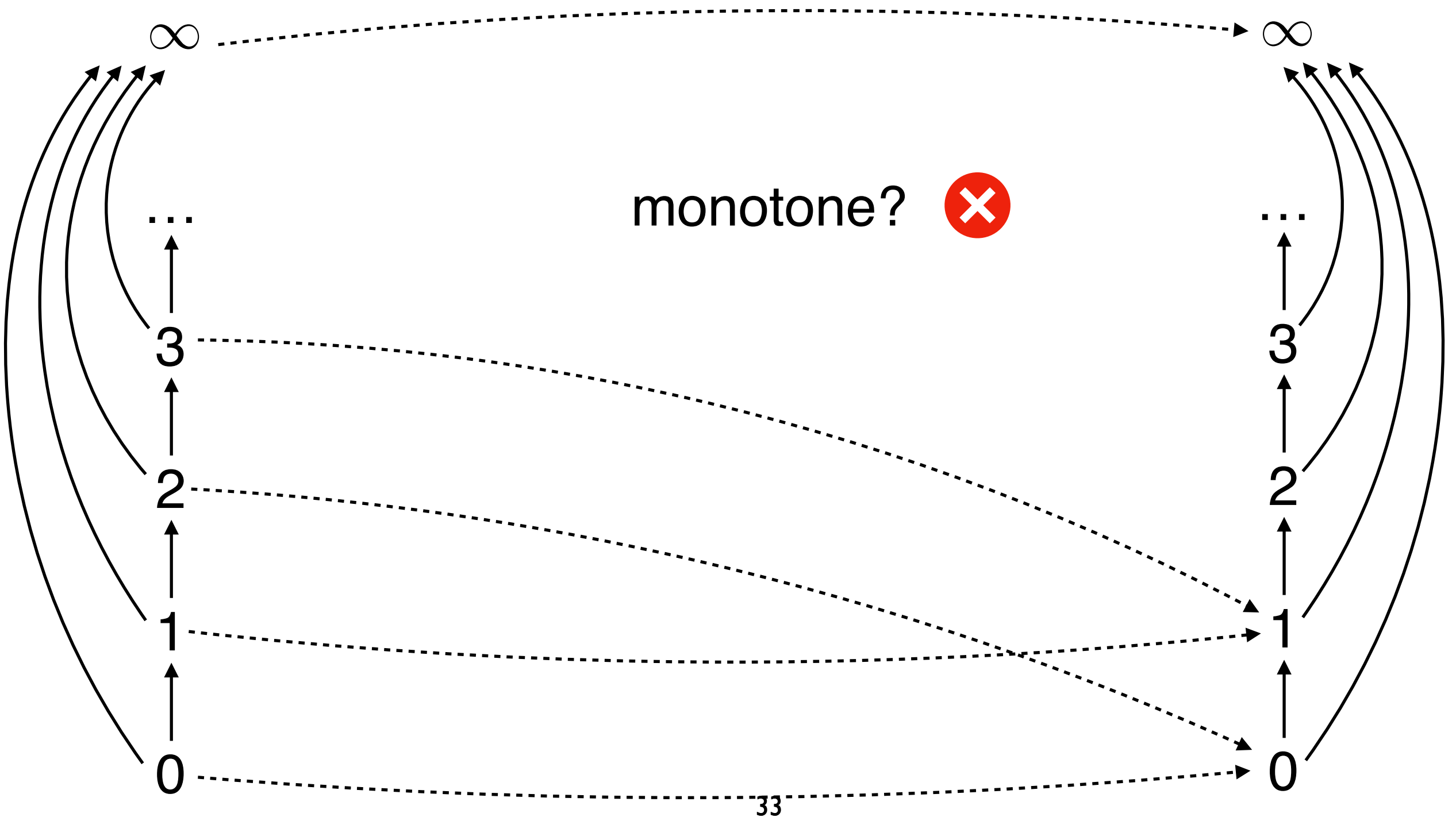
Exercise

$(\mathbb{N} \cup \{\infty\}, \leq)$

$$f(n) = n \% 2$$
$$f(\infty) = \infty$$

$(\mathbb{N} \cup \{\infty\}, \leq)$

monotone?



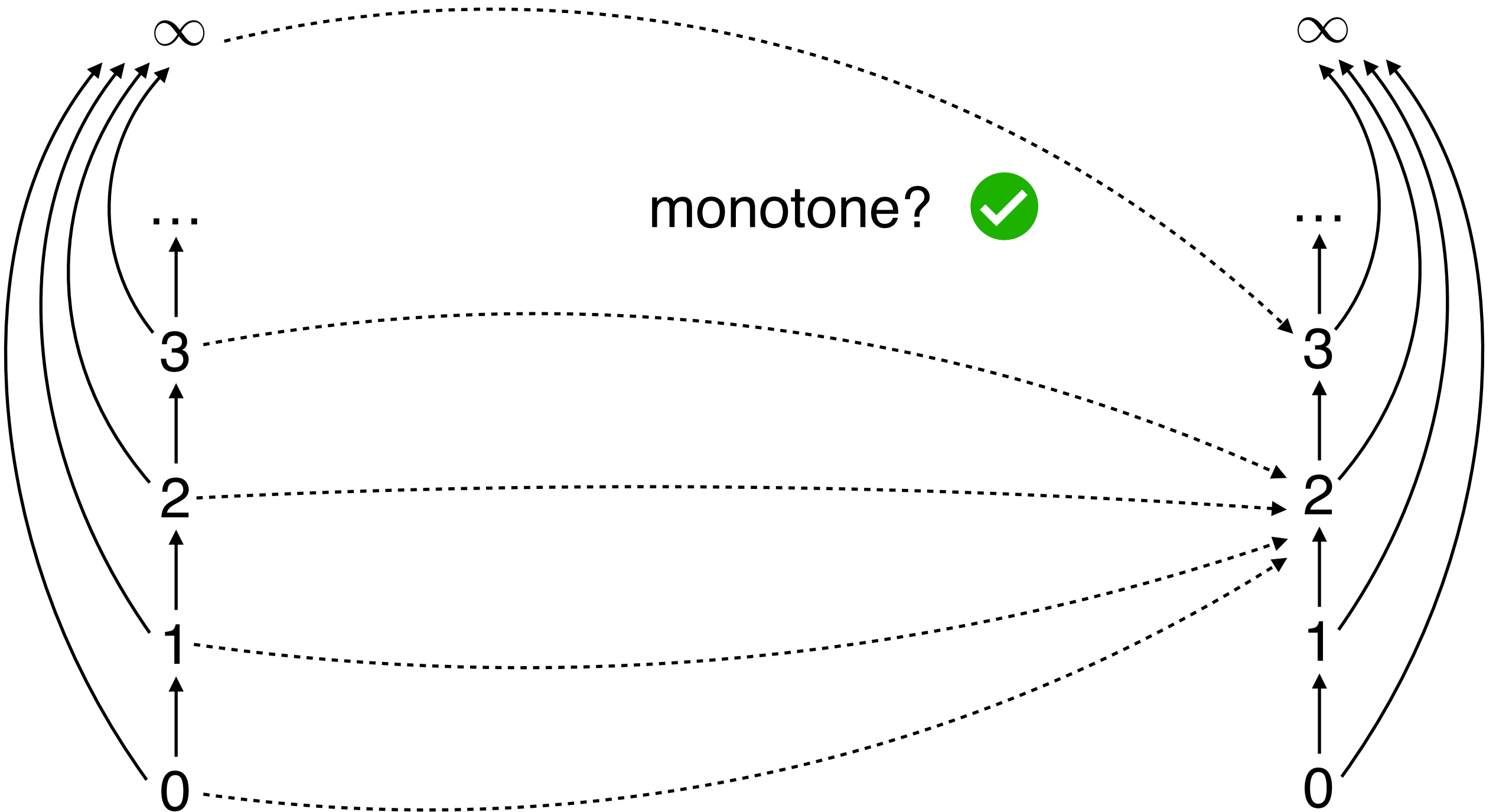


Exercise

$(\mathbb{N} \cup \{\infty\}, \leq)$

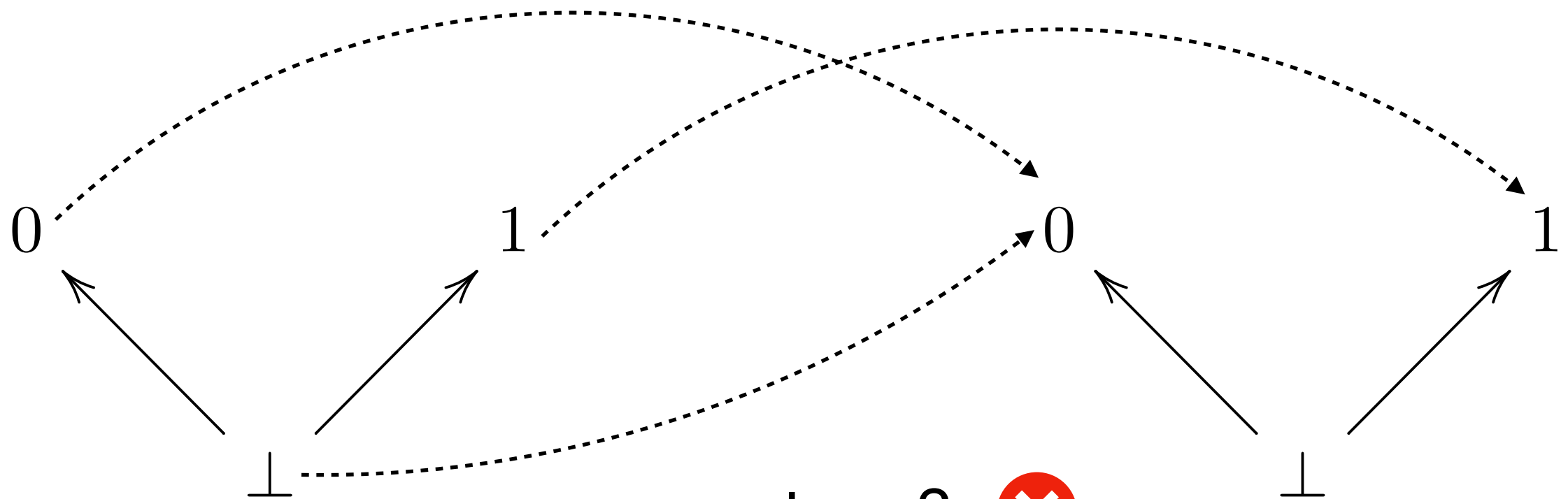
$$\begin{aligned} f(n) &= 2 \\ f(\infty) &= 3 \end{aligned}$$

$(\mathbb{N} \cup \{\infty\}, \leq)$





Exercise



monotone? 

D

$f : D \rightarrow D$

D

$$f(\perp) = f(0) = 0$$

$$f(1) = 1$$

$$\perp \sqsubseteq 1$$

$$f(\perp) = 0 \not\sqsubseteq 1 = f(1)$$



Exercise

$$(\wp(\mathbb{N}), \subseteq) \quad f(S) = \{ m \in \mathbb{N} \mid \exists n \in S, m \leq n \} \quad (\wp(\mathbb{N}), \subseteq)$$

monotone?





Exercise

$$(\wp(\mathbb{N}), \subseteq) \quad f(S) = \{ m \in \mathbb{N} \mid \forall n \in S, n < m \} \quad (\wp(\mathbb{N}), \subseteq)$$

monotone?



Composition

TH. Any composition of monotone functions is monotone

$$\begin{array}{llll} (D, \sqsubseteq_D) & \text{PO} & f : D \rightarrow E \text{ monotone} & \\ (E, \sqsubseteq_E) & \text{PO} & g : E \rightarrow F \text{ monotone} & \Rightarrow h = g \circ f : D \rightarrow F \\ (F, \sqsubseteq_F) & \text{PO} & & \text{monotone} \end{array}$$

proof. we need to prove $\forall x, y \in D. x \sqsubseteq_D y \Rightarrow h(x) \sqsubseteq_F h(y)$

take $x \sqsubseteq_D y$

we want to prove $h(x) \sqsubseteq_F h(y)$

then $f(x) \sqsubseteq_E f(y)$ because f is monotone

then $g(f(x)) \sqsubseteq_F g(f(y))$ because g is monotone

$$\begin{array}{ccc} = & & = \\ h(x) & & h(y) \end{array}$$

Continuous functions

Continuous function

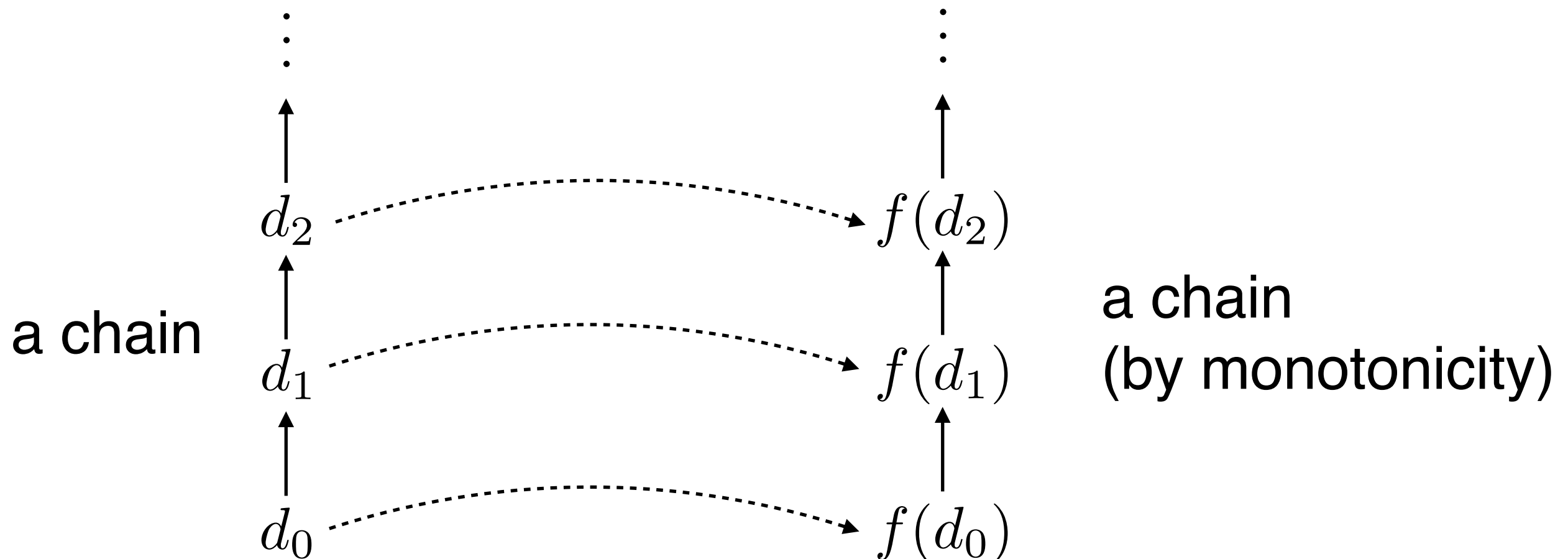
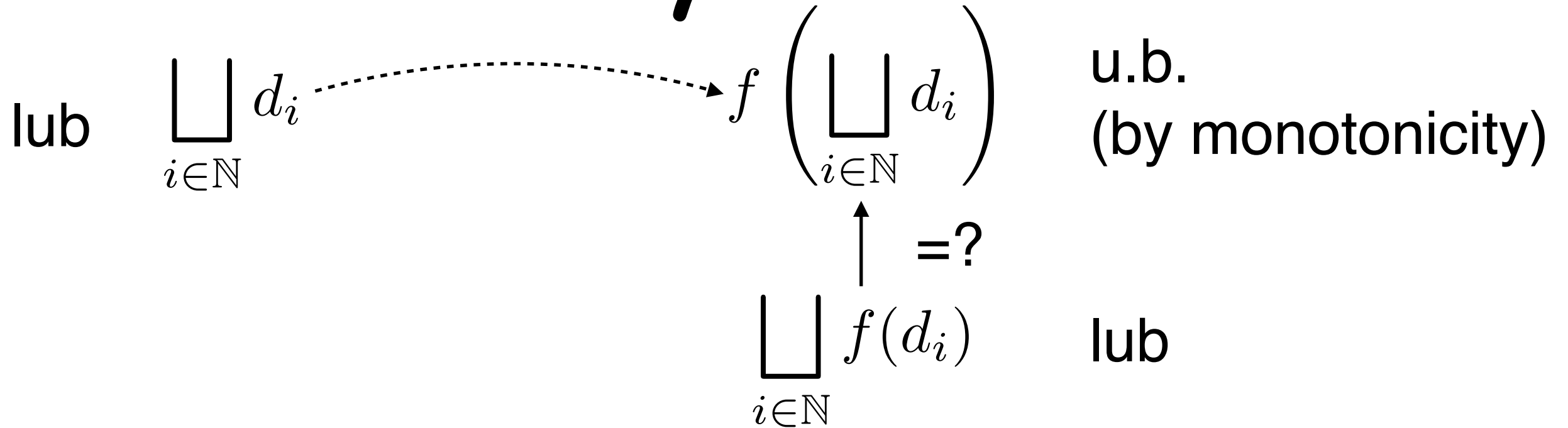
$(D, \sqsubseteq_D)$ CPO $(E, \sqsubseteq_E)$ CPO $f : D \rightarrow E$ monotone

f is **continuous** if $\forall \{d_i\}_{i \in \mathbb{N}} \text{ chain}$ $f \left(\bigsqcup_{i \in \mathbb{N}} d_i \right) = \bigsqcup_{i \in \mathbb{N}} f(d_i)$

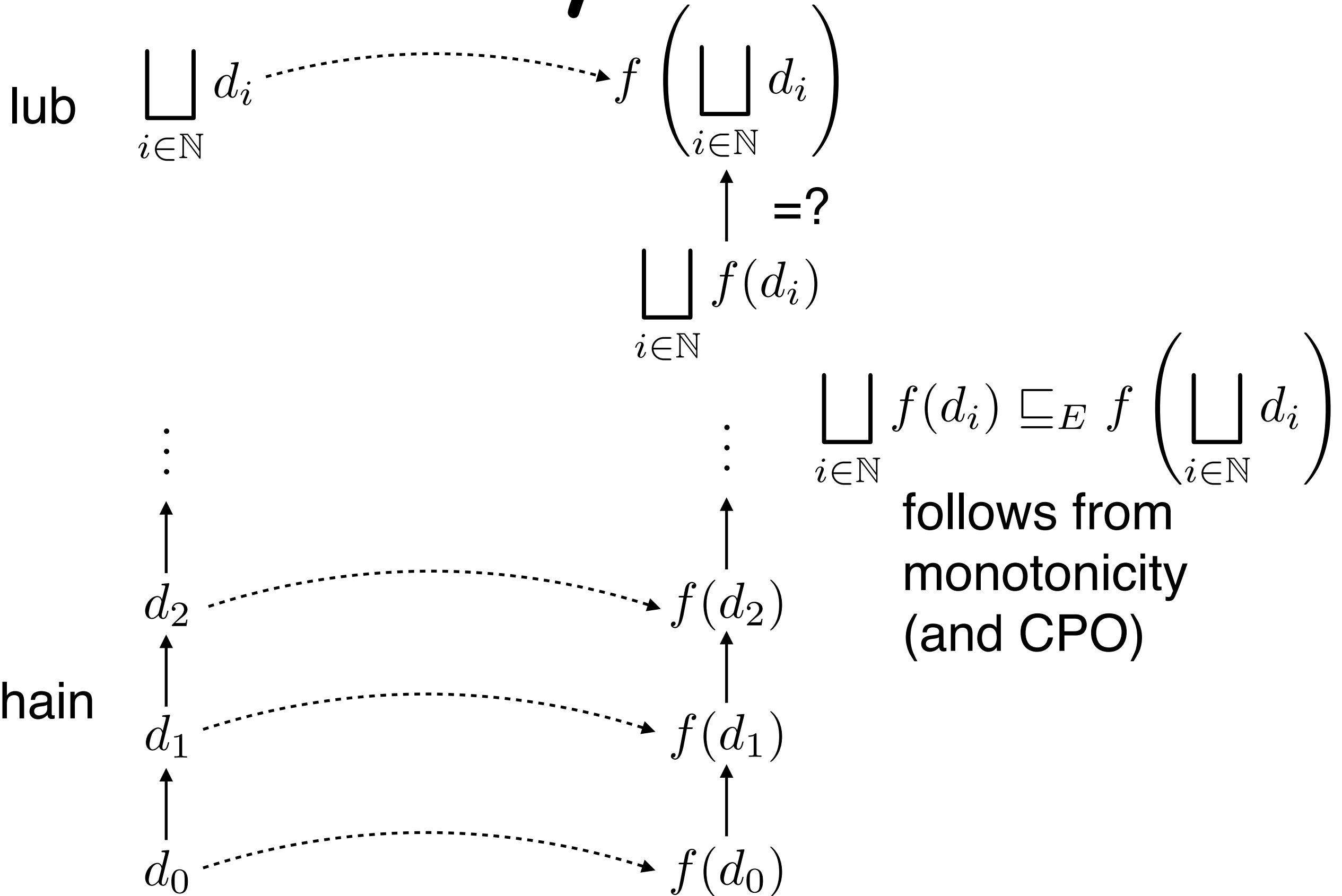
$\text{limit in } D$ $\text{limit in } E$

Continuous = limit preserving

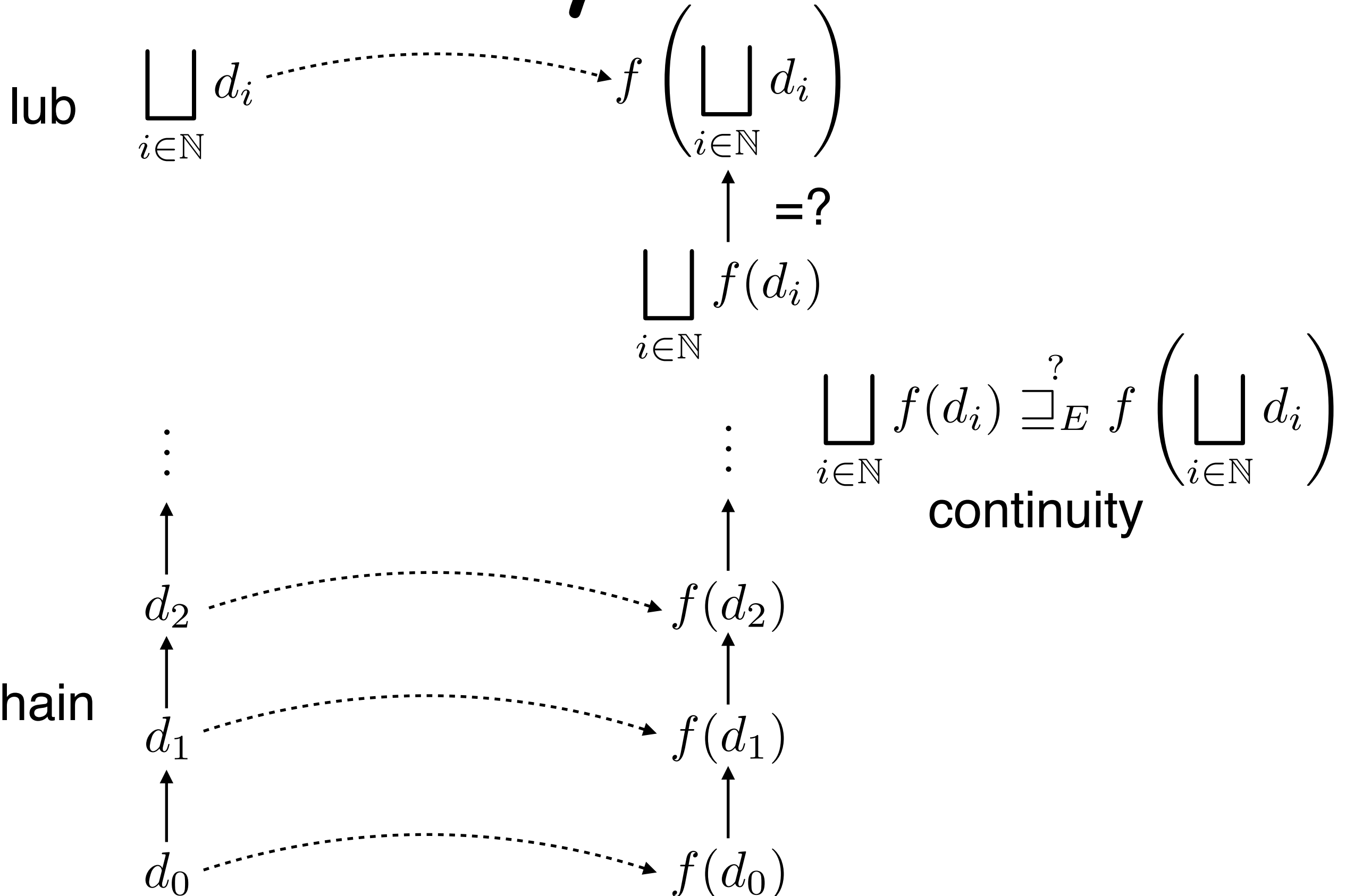
Continuity illustrated



Continuity illustrated



Continuity illustrated

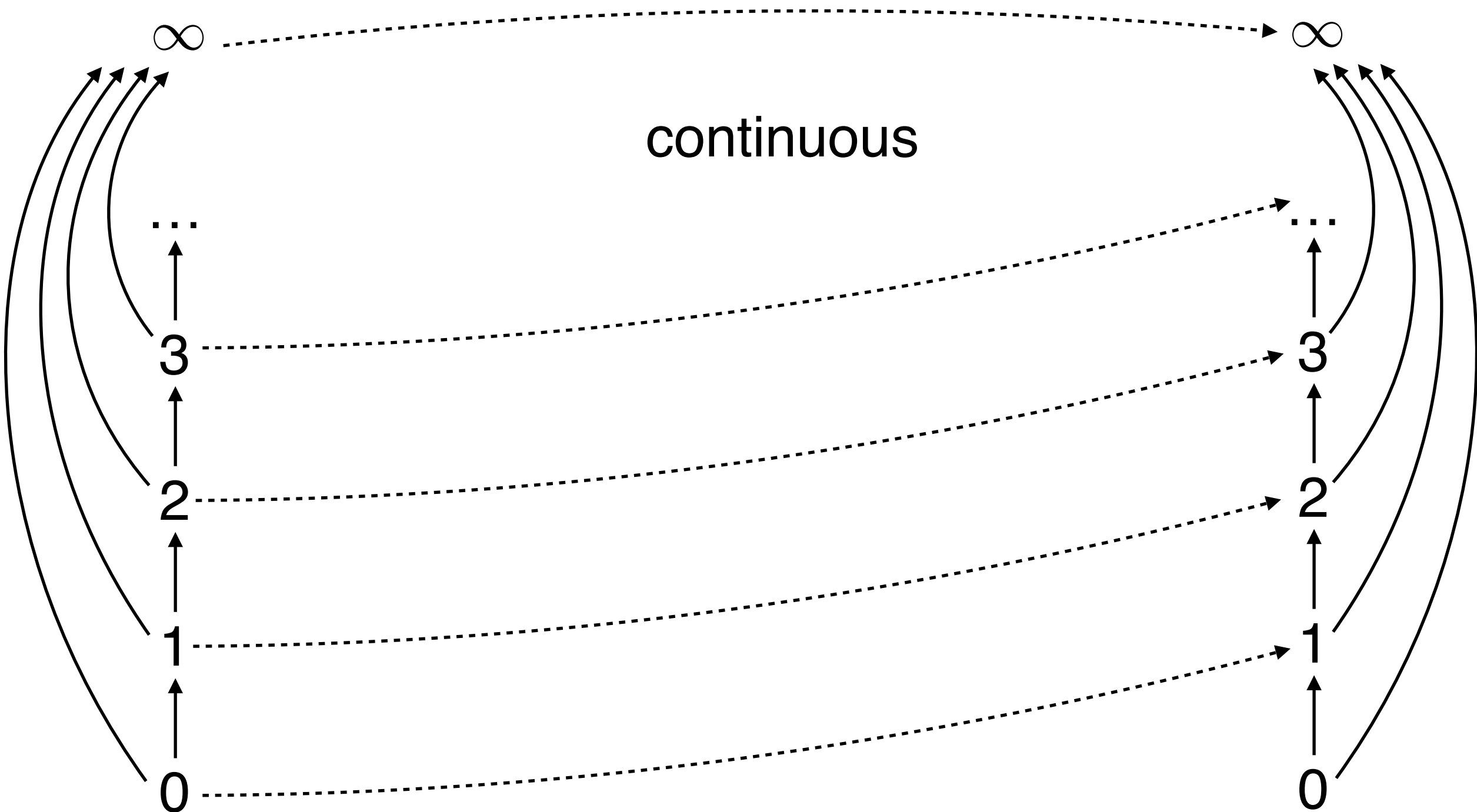


Example

$(\mathbb{N} \cup \{\infty\}, \leq)$

$$f(n) = n + 1$$
$$f(\infty) = \infty$$

$(\mathbb{N} \cup \{\infty\}, \leq)$





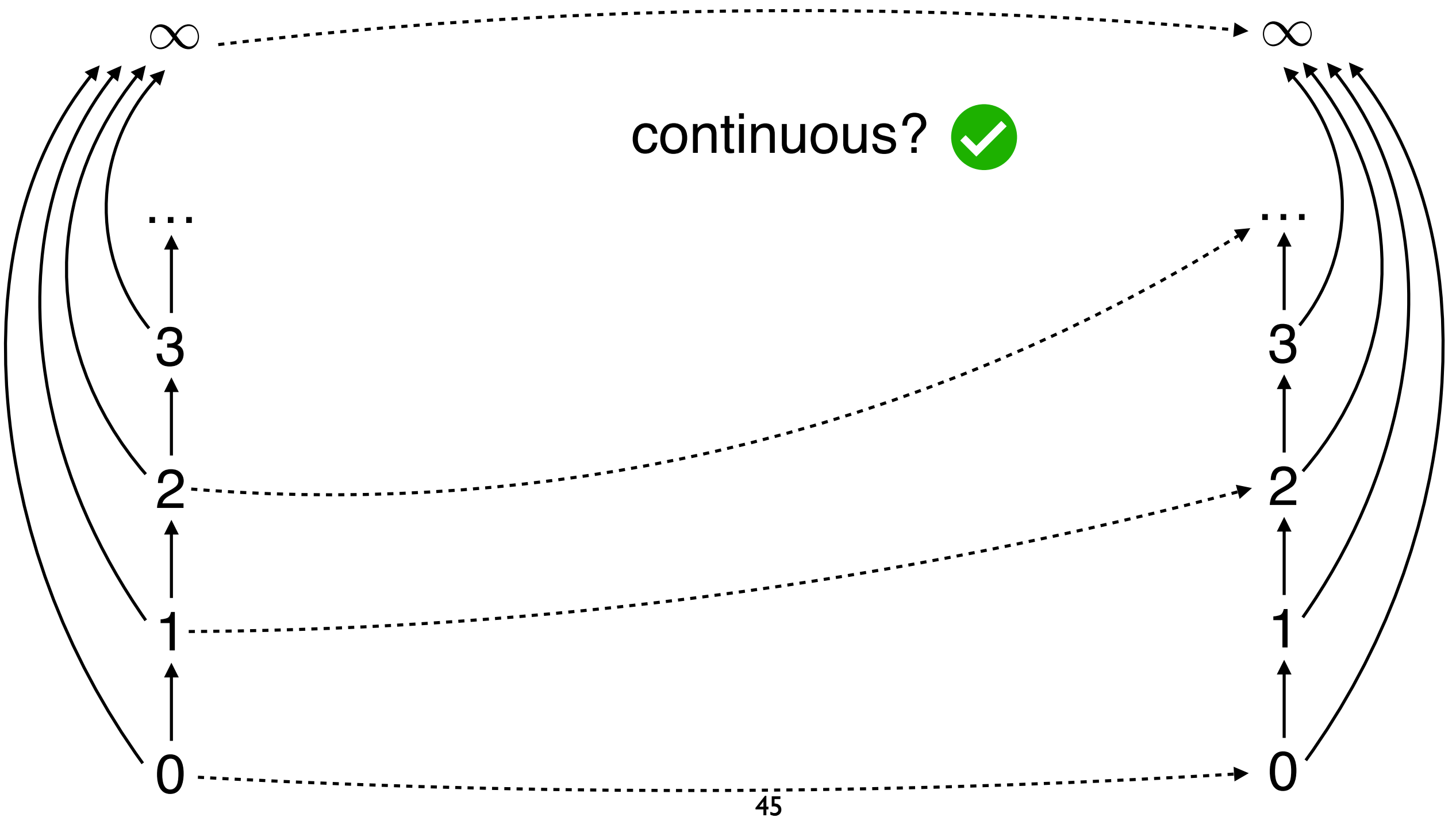
Exercise

$(\mathbb{N} \cup \{\infty\}, \leq)$

$$f(n) = 2 \cdot n$$
$$f(\infty) = \infty$$

$(\mathbb{N} \cup \{\infty\}, \leq)$

continuous? 



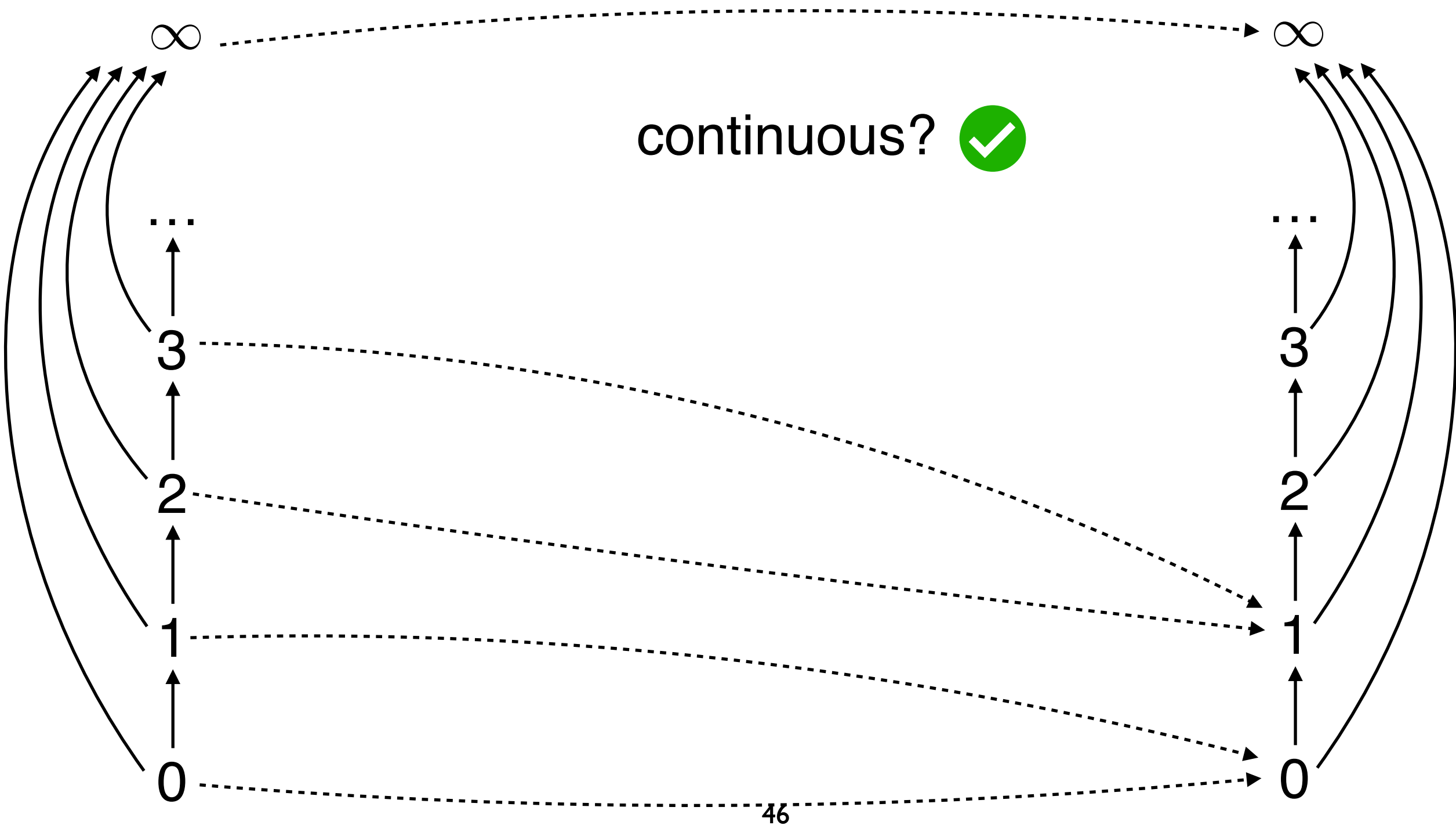
Example

$(\mathbb{N} \cup \{\infty\}, \leq)$

$$f(n) = n/2$$
$$f(\infty) = \infty$$

$(\mathbb{N} \cup \{\infty\}, \leq)$

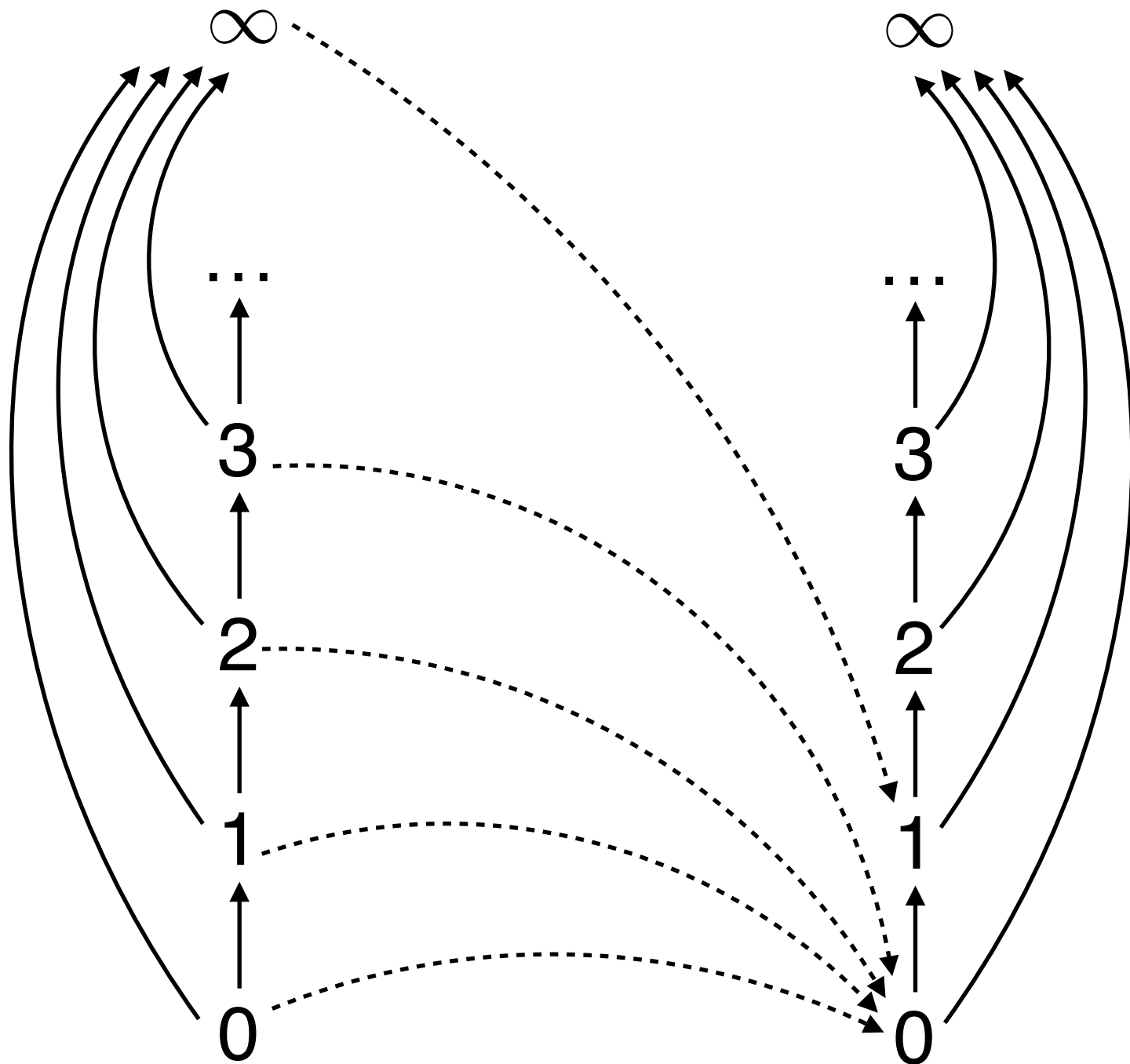
continuous? 



Example

$(\mathbb{N} \cup \{\infty\}, \leq)$

monotone function, not continuous



$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{N} \\ 1 & \text{if } x = \infty \end{cases}$$

$$d_i = 2 \cdot i \quad \bigsqcup_{i \in \mathbb{N}} d_i = \infty$$

$$f\left(\bigsqcup_{i \in \mathbb{N}} d_i\right) = f(\infty) = 1$$

$$f(d_i) = 0$$

$$\bigsqcup_{i \in \mathbb{N}} f(d_i) = \bigsqcup_{i \in \mathbb{N}} 0 = 0$$



Exercise

$$(\wp(\mathbb{N}), \subseteq) \quad f(S) = \{ m \in \mathbb{N} \mid \exists n \in S, m \leq n \} \quad (\wp(\mathbb{N}), \subseteq)$$

continuous? 



Exercise

$(\wp(\mathbb{N}), \subseteq)$

$$f(S) = \begin{cases} S & \text{if } S \text{ finite} \\ \mathbb{N} & \text{otherwise} \end{cases}$$

$(\wp(\mathbb{N}), \subseteq)$

continuous? 

Lemma

$$\begin{array}{llll} (D, \sqsubseteq_D) & \text{CPO} & & \\ \text{no infinite chains} & & f : D \rightarrow E & \Rightarrow f \\ & & \text{monotone} & \text{continuous} \\ (E, \sqsubseteq_E) & \text{PO} & & \end{array}$$

proof. Take a chain $\{d_i\}_{i \in \mathbb{N}}$

$$\{d_i\}_{i \in \mathbb{N}} \text{ is finite} \quad \Rightarrow \quad \exists k \in \mathbb{N}. \bigsqcup_{i \in \mathbb{N}} d_i = d_k$$

$\Downarrow$

$$\{f(d_i)\}_{i \in \mathbb{N}} \text{ is finite} \quad \Rightarrow \quad \bigsqcup_{i \in \mathbb{N}} f(d_i) = f(d_k) = f\left(\bigsqcup_{i \in \mathbb{N}} d_i\right)$$

Composition

TH. Any composition of continuous functions is continuous

$$\begin{array}{lcl}
 (D, \sqsubseteq_D) \text{ CPO} & f : D \rightarrow E \text{ continuous} & \\
 (E, \sqsubseteq_E) \text{ CPO} & g : E \rightarrow F \text{ continuous} & \Rightarrow h = g \circ f : D \rightarrow F \\
 (F, \sqsubseteq_F) \text{ CPO} & & \text{continuous}
 \end{array}$$

proof. take a chain $\{d_i\}_{i \in \mathbb{N}}$

we need to prove

$$h \left(\bigsqcup_{i \in \mathbb{N}} d_i \right) = \bigsqcup_{i \in \mathbb{N}} h(d_i)$$

$$\begin{aligned}
 h \left(\bigsqcup_{i \in \mathbb{N}} d_i \right) &= g \left(f \left(\bigsqcup_{i \in \mathbb{N}} d_i \right) \right) = g \left(\bigsqcup_{i \in \mathbb{N}} f(d_i) \right) = \bigsqcup_{i \in \mathbb{N}} g(f(d_i)) \\
 &= \bigsqcup_{i \in \mathbb{N}} h(d_i)
 \end{aligned}$$

Kleene's fixpoint theorem

Repeated application

$$f : D \rightarrow D$$

$$f^0(d) \triangleq d$$

$$f^{n+1}(d) \triangleq f(f^n(d))$$

$$f^n(d) = \overbrace{f(\cdots (f(d)) \cdots)}^{n \text{ times}}$$

$$f^n : D \rightarrow D$$

Lemma

$(D, \sqsubseteq)$ $\text{PO}_\perp$ $f : D \rightarrow D$ monotone $\Rightarrow$ $\{f^n(\perp)\}_{n \in \mathbb{N}}$
is a chain

proof. we need to prove $\forall n \in \mathbb{N}. f^n(\perp) \sqsubseteq f^{n+1}(\perp)$
by mathematical induction $P(n) \triangleq f^n(\perp) \sqsubseteq f^{n+1}(\perp)$

$$P(0) \triangleq f^0(\perp) \sqsubseteq f^1(\perp) \qquad f^0(\perp) = \perp \sqsubseteq f^1(\perp)$$

$\forall n \in \mathbb{N}. P(n) \Rightarrow P(n+1)$ take a generic n

assume $P(n) \triangleq f^n(\perp) \sqsubseteq f^{n+1}(\perp)$

we want to prove $P(n+1) \triangleq f^{n+1}(\perp) \sqsubseteq f^{n+2}(\perp)$

$$\begin{array}{c} f^n(\perp) \sqsubseteq f^{n+1}(\perp) \\ \Downarrow \\ f^{n+1}(\perp) = f(f^n(\perp)) \sqsubseteq f(f^{n+1}(\perp)) = f^{n+2}(\perp) \end{array}$$

Towards Kleene's theo.

when $(D, \sqsubseteq)$ is a $\text{CPO}_\perp$

then $\{f^n(\perp)\}_{n \in \mathbb{N}}$ is a chain

it must have a limit

$\{f^n(d)\}_{n \in \mathbb{N}}$
not necessarily
a chain!

Kleene's fix point theorem states that
if f is continuous, then the limit of the above chain
is the least fixpoint of f

Pre-fixpoints

$(D, \sqsubseteq)$ PO $f : D \rightarrow D$ monotone

fixpoint $p \in D$ $f(p) = p$

pre-fixpoint $p \in D$ $f(p) \sqsubseteq p$

Clearly any fixpoint is also a pre-fixpoint

Kleene's theorem

$(D, \sqsubseteq)$ CPO $_{\perp}$ $f : D \rightarrow D$ continuous

let $fix(f) \triangleq \bigsqcup_{n \in \mathbb{N}} f^n(\perp)$

1. $fix(f)$ is a fix point of f

$$f(fix(f)) = fix(f)$$

2. $fix(f)$ is the least pre-fixpoint of f

$$\forall d \in D. f(d) \sqsubseteq d \Rightarrow fix(f) \sqsubseteq d$$

if d is a pre-fixpoint then $fix(f)$ is smaller than d

Kleene's theorem: 1

$$1. \quad f(\text{fix}(f)) = \text{fix}(f)$$

proof.

$$f(\text{fix}(f)) = f\left(\bigsqcup_{n \in \mathbb{N}} f^n(\perp)\right) \quad \text{by def of } \text{fix}$$

$$= \bigsqcup_{n \in \mathbb{N}} f(f^n(\perp)) \quad \text{by continuity}$$

$$= \bigsqcup_{n \in \mathbb{N}} f^{n+1}(\perp) \quad \text{by def of } f^n$$

$$= \bigsqcup_{n \in \mathbb{N}} f^n(\perp) \quad \text{by prefix independence of limits}$$

$$= \text{fix}(f) \quad \text{by def of } \text{fix}$$

Kleene's theorem: 2

$$2. \quad \forall d \in D. f(d) \sqsubseteq d \Rightarrow \text{fix}(f) \sqsubseteq d$$

proof.

we prove that any pre-fixpoint is an upper bound of the chain

$$\{f^n(\perp)\}_{n \in \mathbb{N}}$$

by definition $\text{fix}(f)$ is the lub of the same chain
and thus smaller than any other upper bound

Kleene's theorem: 2

$$2. \quad \forall d \in D. f(d) \sqsubseteq d \Rightarrow \text{fix}(f) \sqsubseteq d$$

take any $d \in D$ such that $f(d) \sqsubseteq d$

we prove $\forall n \in \mathbb{N}. f^n(\perp) \sqsubseteq d$ (d is an upper bound)

$P(n) \triangleq f^n(\perp) \sqsubseteq d$ by mathematical induction

$$P(0) \triangleq f^0(\perp) \sqsubseteq d \quad f^0(\perp) = \perp \sqsubseteq d$$

$\forall n \in \mathbb{N}. P(n) \Rightarrow P(n+1)$ take a generic n
assume $P(n) \triangleq f^n(\perp) \sqsubseteq d$

we want to prove $P(n+1) \triangleq f^{n+1}(\perp) \sqsubseteq d$

$$\begin{array}{ccccc} & & f^n(\perp) \sqsubseteq d & & \\ & & \downarrow \text{(monot.)} & & \\ f^{n+1}(\perp) & \stackrel{\text{(by def)}}{=} & f(f^n(\perp)) & \sqsubseteq & f(d) \sqsubseteq d \quad \text{(pre-fixpoint)} \end{array}$$

Example

$$n = 2 \cdot n$$

$$(\mathbb{N} \cup \{\infty\}, \leq)$$

$$\perp = 0$$

$\text{CPO}_{\perp}$

$$\begin{aligned} f(n) &= 2 \cdot n \\ f(\infty) &= \infty \end{aligned}$$

monotone? ok

continuous? ok

$$f^0(0) = 0$$

$$f^1(0) = f(0) = 2 \cdot 0 = 0$$

fixpoint reached!

Example

$$n = n + 1$$

$$(\mathbb{N} \cup \{\infty\}, \leq)$$

$$\perp = 0$$

$\text{CPO}_{\perp}$

$$\begin{aligned} f(n) &= n + 1 \\ f(\infty) &= \infty \end{aligned}$$

monotone? ok

continuous? ok

$$f^0(0) = 0$$

$$f^1(0) = f(0) = 0 + 1 = 1$$

$$f^2(0) = f(f^1(0)) = f(1) = 1 + 1 = 2$$

$$f^3(0) = f(f^2(0)) = f(2) = 2 + 1 = 3$$

$$\bigsqcup_{n \in \mathbb{N}} f^n(0) = \bigsqcup_{n \in \mathbb{N}} n = \infty \quad \text{fixpoint}$$

Example

$$X = X \cap \{1\}$$

$$(\wp(\mathbb{N}), \subseteq)$$

$$\perp = \emptyset \quad \text{CPO}_{\perp}$$

$$f(X) = X \cap \{1\}$$

monotone? ok
continuous? ok

$$f^0(\emptyset) = \emptyset$$

$$f^1(\emptyset) = f(\emptyset) = \emptyset \cap \{1\} = \emptyset$$

fixpoint reached!

Example

$$X = \mathbb{N} \setminus X$$

$$(\wp(\mathbb{N}), \subseteq)$$

$$\perp = \emptyset \quad \text{CPO}_{\perp}$$

$$f(X) = \mathbb{N} \setminus X$$

monotone? NO

the larger X the smaller $f(X)$

$$f^0(\emptyset) = \emptyset$$

$$f^1(\emptyset) = f(\emptyset) = \mathbb{N} \setminus \emptyset = \mathbb{N}$$

$$f^2(\emptyset) = f(f^1(\emptyset)) = f(\mathbb{N}) = \mathbb{N} \setminus \mathbb{N} = \emptyset$$

not a chain!

Example

$$X = X \cup \{1\}$$

$$(\wp(\mathbb{N}), \subseteq)$$

$$\perp = \emptyset \quad \text{CPO}_{\perp}$$

$$f(X) = X \cup \{1\}$$

monotone? ok
continuous? ok

$$f^0(\emptyset) = \emptyset$$

$$f^1(\emptyset) = f(\emptyset) = \emptyset \cup \{1\} = \{1\}$$

$$f^2(\emptyset) = f(f^1(\emptyset)) = f(\{1\}) = \{1\} \cup \{1\} = \{1\}$$

fixpoint reached!

Badge exercise



Let D be a CPO

let $\{d_i\}_{i \in \mathbb{N}}$ be a chain in D

let $\{k_j\}_{j \in \mathbb{N}}$ be an infinite chain in $(\mathbb{N}, \leq)$

1. Prove that $\{d_{k_j}\}_{j \in \mathbb{N}}$ is a chain in D

2. Prove or disprove that $\bigsqcup_{j \in \mathbb{N}} d_{k_j} = \bigsqcup_{i \in \mathbb{N}} d_i$